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Overview

For a video overview of Work Release, visit the Warehouse Management Video Training page.
Work Release is a key feature in the order fulfillment process that is responsible for releasing work to the warehouse floor by evaluating the availability of equipment, labor, and inventory. It prioritizes critical orders and aims to optimize resource utilization and throughput. To tailor release decisions to specific business needs, users can define appropriate business rules that guide the process.

Distribution Orders intended for release can be planned using either order streaming or a wave strategy. In both strategies, orders are assigned to a pipeline - a defined sequence of steps an order goes through - and are automatically selected for fulfillment.

- In the order streaming approach, multiple orders are batched for allocation and then proceed through a series of fulfillment steps. These steps may include cubing and routing, both of which are optional. Once the orders reach the work release stage, additional steps such as order consolidation and printing may be performed if required.
- A wave strategy involves selecting a group of orders that move through a predefined wave pipeline. This pipeline typically includes allocation, cubing, routing, order consolidation, and work release. While allocation and work release are mandatory, the other steps are optional and can be configured based on operational needs. Printing, if necessary, occurs after the work release step.

Work release serves as a control center that continuously monitors and identifies high-priority work, creates efficient tasks and releases them for execution. It maintains a real-time view of outbound equipment capacity, such as unit sorters, putwalls, and pack stations, alongside current inventory levels and pending work. By doing so, it ensures that the most suitable tasks are released to the floor, aligning with the warehouse's fulfillment strategy and operational constraints.

The work release process is driven by two main engines:

- Capacity Manager: The Capacity Manager is responsible for identifying the most important work, assigning equipment resources based on capacity and inventory availability, and distribution of pick allocations to pack allocations in case of bulk flow.
- Task Release Manager: The Task Release Manager then creates efficient tasks from the selected work and releases them according to equipment capacity and labor constraints.

Together, these components ensure that task execution is both timely and resource-optimized.

The frequency of the work release process is controlled through the "`workReleaseSchedulerJobSchedule`". This configuration allows users to define how often the base flow of work release should be invoked, ensuring that task release aligns with operational rhythms and throughput goals.

Work release is the final step in the order orchestration pipeline, marking the shift from planning to execution. It typically follows sequences such as:

1. Allocation > Routing > Work Release
2. Allocation > Cubing > Routing > Work Release

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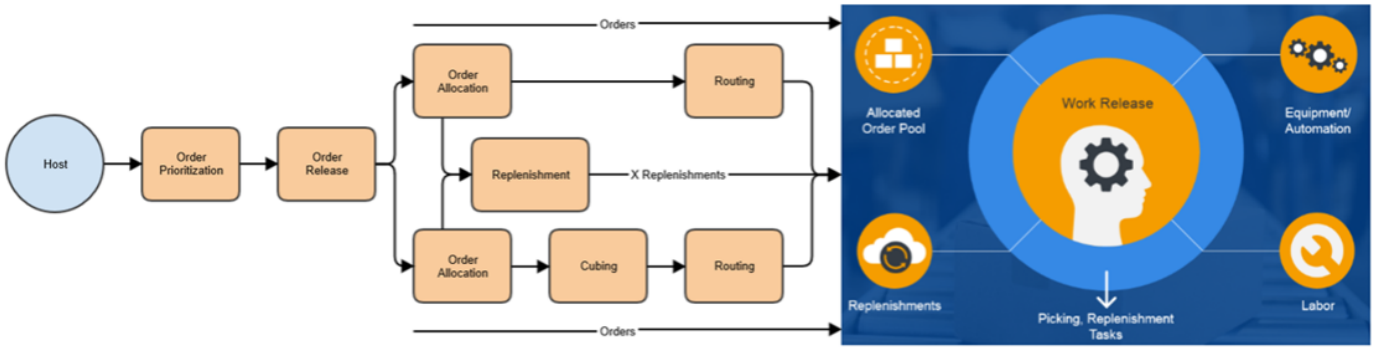


What is 'Work'

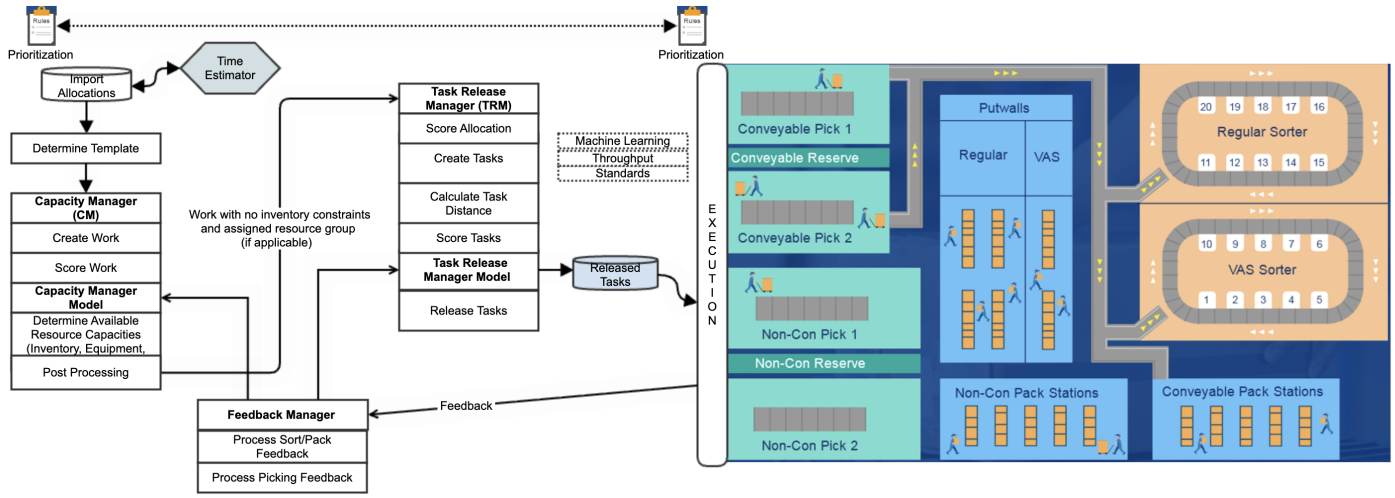
In the context of work release, Work is a fundamental unit used to manage and allocate equipment capacity. It represents the smallest grouping of allocations required to match available inventory with resource capacity. Each unit of work occupies one unit of capacity on the assigned resource. A resource refers to any equipment that processes work, typically a putwall, unit sorter, chute, or a pack station. When an allocation enters the work release process, it is assigned a set of work release criteria. These criteria determine how the work is created, how it is scored, and which resource groups are eligible to process it.

> **Note**: In this document, the terms "resource" and "equipment" are used interchangeably.

Resources are systemically grouped into resource groups based on the type of work they are designed to process. One of the primary functions of the work release process is to determine the appropriate resource group and, if configured, the specific resource within that group that should process the given unit of work.

Example: Depending on how the resource capacity is measured, work should be created at an appropriate level. Consider a sorter chute. Its capacity is measured in number of oLPN per chute (maximum of 5 *oLPN* per chute). In such a scenario, the work creation level should be oLPN. However, if the capacity is 3 *orders* per chute, then the work creation level should be order. A group of chutes form a resource group (such as regular chutes and VAS chutes). Work release will typically manage capacity at a resource group level but, if required, a logical resource can be assigned. For retail flows, pick and replenishment allocations, work creation level should be allocation.

Process Flow



Direct Task Creation

Direct Task Creation allows the tasks for a wave to be created and released immediately upon wave completion, bypassing the intermediate constraint checks like inventory, resource capacity, or labor capacity. The advantage of Direct Task Creation is the

reduced time between the wave completion and task release, allowing the tasks to be assigned to pickers quickly for operations that do not want to use any constraints. This is supported only for non-bulk flow. Direct Task Creation also makes sure all the allocations of the wave are processed together for task creation. This can be enabled in the Picking Task Creation Strategy configuration. If Direct Task Creation is enabled, it is recommended not to enable any constraint checks or minimum task thresholds.

The table below lists the relevant criteria assigned to an allocation that is processed in the Direct Task Creation flow. The Work Release Template will not be assigned for allocations processed in this flow.

Allocation Type	Relevant Strategy	Relevant Criteria
PICK/PACK	Picking Task Strategy	Task Criteria
REPLENISHMENT	Replenishment Task Strategy	Task Criteria

To enable Direct Task Creation for replenishment-only waves, enable the Direct Task Creation parameter in the Replenishment Task Strategy. To enable Direct Task Creation for lean time replenishments:

- Enable the Direct Task Creation parameter in Replenishment Task Strategy.
- Set the 'delayExportLeanTimeReplenAllocations' to True in the Allocation API.
- Set the 'Enable Lean Time Visibility' attribute in Allocation Parameters to 'True'.

In Direct Task Creation, all allocations for a wave get processed together and hence, if there are a few allocations that do not get assigned to a template, the entire wave's allocations will be held back without releasing. In such scenarios, flexibility is available to decide whether to hold back all allocations or release the eligible allocations based on a configurable threshold. This provides greater flexibility in handling the template assignment failed allocations while ensuring that the eligible ones can still proceed for task creation and release. The allocation failure threshold in the Picking Task Creation Strategy defines the maximum percentage of template assignment failed allocations to allow release of eligible allocations. If the percentage of the template assignment failed allocations within a wave is equal to or falls below the threshold configured, the eligible allocations will be released by ignoring the failures.

The default behavior for Direct Task Creation is to bypass the failed template assignment allocations and release the eligible allocations within the wave.

If the percentage of template assignment failed allocations within the wave is beyond the configured threshold, the users need to troubleshoot the configuration issue and fix it. If the configuration issues are fixed within a 15-minute window, there is no manual action required. If it is beyond the 15-minute window, the users can navigate to the Wave Runs UI and take the 'Generate Task' action to:

- Force-assign a single template to all the failed allocations within the wave
- Trigger the template re-assignment process
- Trigger the template re-assignment process and force-assign a single template to the remaining
- Ignore the failed allocations and force-release the eligible allocations within the wave

Import Allocations

Allocations are the foundational units in the work release process. When orders are ready for processing, DCO (DC Order) notifies work release of their availability. Work release then pulls these allocations for further processing.

The following types of allocations can exist within work release:

1. PICK/PACK - A non-bulk allocation indicating that the inventory (item, location, quantity) allocated is already committed to a

specific order or oLPN.

2. PICK, SINGLES_PICK - A bulk allocation used to fulfill aggregate demand across multiple orders. These allocations are not tied to any specific order.
3. PACK - A bulk allocation where the order is not yet committed to a specific oLPN.
4. REPLENISHMENT - An allocation type used to move inventory within the warehouse to replenish depleted locations.

Process Flow

1. DC Order orchestrates the fulfillment of Order/Wave and notifies work release when an Order + Order Plan Run ID (in case of Order Streaming) or Order Plan Run ID (in case of Wave) is ready for work release.
2. Work release processes this information and pulls all allocations tied to the above notification from Allocation Store (DC Inventory).
3. After allocations have been successfully imported, work release sends an acknowledgment to DC Order.
4. Allocations are available for further processing only after the entire set tied to the notification is received by Work Release.
5. The "ExportPickAndReplenAllocations" flag in DC Order Parameters controls how the allocations are exported to work release. By default, this flag is set to 'False', which means the work release component pulls the allocations for the wave during the import allocation process. If the flag is set to 'True' or NULL, the allocations are "pushed" into work release. Similarly, to pull the lean time replenishment allocations, the "DelayExportingLeanTimeReplenishmentAllocations" flag should be set to 'True' and the Lean Time Visibility parameter should be enabled in the Allocation Parameters.

Allocation Type	Trigger	Action
PICK/PACK PACK	DCO puts a message in the Work Release request queue for an order plan run ID and/or order ID combination.	Work Release uses this message to pull the allocation from DC Inventory.
PICK SINGLES_PICK REPLENISHMENT DEFERRED_REPLENISHMENT DEFERRED_PICK_PACK	The Allocation component publishes the allocations to a queue.	Work release consumes this message and pulls the allocation from DCA (DC Allocation).

Business Updates

Allocations are persisted in the "allocation" entity of work release. The status of the imported allocation is:

Status	Allocation Type
READY_FOR_WORK_RELEASE	PICK/PACK PICK, SINGLE_PICK, PACK, REPLENISHMENT
DEFERRED_FOR_ASSIGNMENT	DEFERRED_PICK_PACK, DEFERRED_REPLENISHMENT

Post persistence of allocations in work release, allocations in the status of READY_FOR_WORK_RELEASE are moved to the status of "AWAITING_TEMPLATE_ASSIGNMENT". The allocations then undergo criteria determination.

Determine Templates

Post allocation import, criteria(s) are determined for allocations. This is done by utilizing the strategy/rule framework, where allocations and corresponding run IDs are input, and the criterion is an output. The criterion carries the configuration information that determines the path an allocation can take in work release.

Process Flow

1. Each allocation is tied to an order plan run ID.
2. For each run ID, DCO communicates the strategies to work release. The exact strategies are selected and configured in the

"Order Planning Strategy". The relevant strategies for work release are:

- Sort Pack Resource Strategy
- Replenishment Resource Strategy
- Picking Task Strategy
- Replenishment Task Strategy

3. The run strategies are stored in the corresponding "RunStrategy" entities in Work Release.
4. For each of the allocations, based on Run ID, strategies are determined. The relevant strategies are dependent on the allocation type. See the matrix below.
5. Based on the rule framework, relevant work release and task criteria are assigned to an allocation.

Allocation Type	Relevant Strategies	Relevant Criteria	Comments
PICK/PACK	Sort Pack Resource Strategy Picking Task Strategy	Work Release Criteria Task Criteria	For allocations tied to an order need in non-bulk flow.
PACK	Sort Pack Resource Strategy	Work Release Criteria	For allocations tied to an order need in bulk flow.
REPLENISHMENT	Replenishment Resource Strategy Replenishment Task Strategy	Work Release Criteria Task Criteria	Replenishment allocations are not tied to specific orders. In case of replenishment for a stream mode, only default replenishment strategy will be used. It is advised to maintain a default strategy for replenishment.
PICK	Picking Task Strategy	Task Criteria	For allocations tied to an order need in bulk flow.

Note: The above table is the recommended use of template strategies for a given set of allocation types. It would result in a failure, if allocation are not appropriately conditioned to their respective strategies.

Configuration

Based on the rules defined in the work release strategies (Sort Pack Resource, Replenishment Resource, Picking Task and Replenishment Task), the work release criteria and task criteria are determined for the allocation.

Work Release Criteria

Work Release criteria determine whether capacities are modeled for relevant allocations, and if so, specify how they are modeled. A sort pack resource strategy and replenishment resource strategy can be associated with one or more work release criteria. These criteria include attributes that define how work is created, scored and whether resources are modeled. The criteria are stored in the "workReleaseTemplate" entity. The entities available for configuring selection rules vary depending on the strategy.

Task Criteria

Task criteria define how allocations are tasked in work release. They include attributes that specify how tasks are created, as well as threshold criteria that influence task creation. Each Picking Task Strategy and Replenishment Task Strategy can be associated with one or more task criteria. These criteria are stored in the "taskReleaseTemplate" entity. The entities available for configuring selection rules for task criteria vary depending on the strategy.

Business Updates

Once all the allocations for an 'Order Plan Run ID/Order ID' have a template assigned, the allocation status moves to

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”TEMPLATE_ASSIGNED". Allocations are updated with the work release and the task criteria (as applicable).

In case of failure to determine work release criteria, the allocation is moved to a status of TEMPLATE_ASSIGNMENT_FAILED. The rules should be adjusted to correct the failure. At the start of the work release scheduler, the failed allocations are picked up and rules are run again to identify the work release criteria.

Capacity Manager

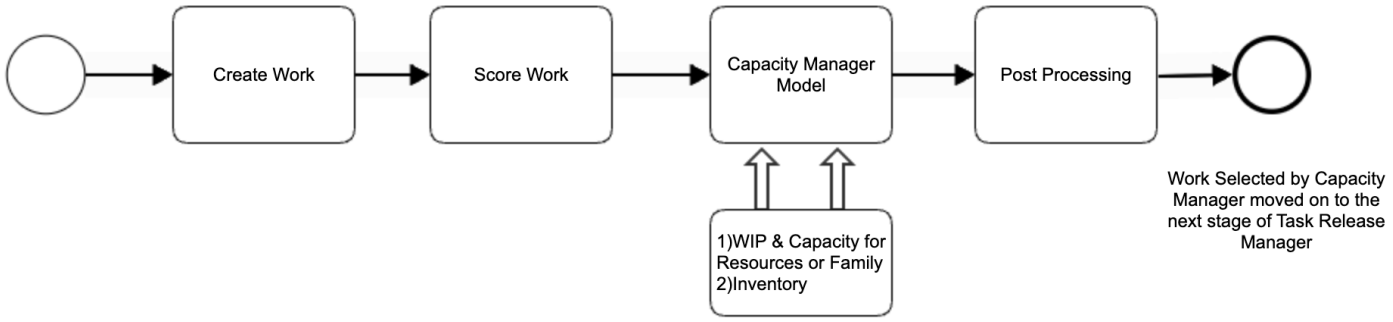
This engine initiates and sets the stage for the entire work release process. It has visibility of the pending pool of work, available capacity of each resource group and the current status of inventory. Using this information, the Capacity Manager identifies the highest-priority work and allocates them to the appropriate resource groups for processing.

Note: The configuration of a resource constraint is optional.

The primary objective of the Capacity Manager is to evaluate the pending pool of work, determine their relative importance, verify inventory availability and assign the highest-priority work to appropriate resource groups or, if configured, to specific work resources. Resource capacities are evaluated at the work UOM level. The following steps outline the capacity manager's workflow:

1. Create Work - Allocations are grouped into an entity called "Work". Each "Work" unit represents the smallest grouping for which inventory and resource capacity constraints (if applicable) are evaluated.
2. Score Work - All work units are scored based on their relative priority and cutoff time.
3. Capacity Manager Model - The capacity manager uses an optimization model to evaluate inventory, resource capacity and work scores. The goal is to select the most important work while maximizing resource utilization. Optionally, this step can also create Resource Group Family batches (Pack Waves).
4. Post Processing - The model's output is consumed and the selected work(s) are advanced to the next status. If configured in work resource group, specific work resource is also assigned.

This structured approach ensures that the most critical work is prioritized and that resources are used efficiently.



Each of the steps mentioned above is explained in detail:

Capacity Manager
Create Work → Score Work→ Capacity Manager Model→ Post-Processing

Create Work

Work is a group of allocations. At the most granular level, a work can be created for the allocation itself. Each allocation is assigned a work criterion.

Work is created in work release based on the allocations and work release criteria. A work ID is generated for the work. The

composition of the work ID can be controlled by the Next Up configuration for “WorkId”. The work criterion carries important attributes about work and has an impact on the work process by work release.

Work Size

Work Size refers to the number of space units that a work occupies at a resource.

Typically, each work record occupies one unit of space in the resource. For example, if the work record is created at the oLPN level and the resource is a putwall with a fixed number of cubbies (e.g., 20), where each cubby holds one oLPN, then the resource has a capacity of 20. In this scenario, the work size is 1, and each oLPN occupies one unit of space.

In certain scenarios, a single work record may occupy multiple units of space at a resource. For example, if the work record is created at the oLPN level and the resource is a tote bank, where each allocation (pick) for an oLPN is placed in a separate tote, then the space consumed by that oLPN depends on the number of allocations. If the tote bank has a capacity of 1,000 totes, and an oLPN has 3 picks, it would occupy 3 totes, therefore consuming 3 units of space.

To enable greater flexibility in modeling work record sizes, we have provided the ability to define each work record size using an expression. This is configured through the Work Capacity Expression attribute in the work release template. Using this feature, the first use case mentioned above can be addressed by setting the work size to a constant value of 1 (work). The second use case can be handled by setting the work size expression to reflect the allocation count for each work record.

Configuration

Work Release Criteria for Sort Pack Resource Strategy

- Work Creation Level — The work creation level should be determined based on how resources are modeled from a capacity standpoint. This allows for flexible configuration depending on operational needs.
  - oLPN-Level Work Creation: If the shippable entity in a distribution center (DC) is oLPN, then resource capacity can be modeled at the oLPN level.
  - Order-Level Work Creation: If the requirement is to pick an entire order together (comprising multiple oLPNs), then work can be created at the order level. This is typically applicable for non-cubed flows.
  - Allocation-Level Work Creation: For retail and replenishment flows, work is created at the allocation level.
  - Pallet-Level Work Creation: If the requirement is to ship or handle an entire pallet together, then work can be created at the pallet level.
- The system supports flexible resource modeling to align with the building's operational needs.
- For PICK/PACK and PACK allocations, the work creation level is configurable and can be set to Order, oLPN, Pallet or Allocation.
- For Pick and Replenishment allocations, work is always created at the Allocation level.
- If inventory and resource checks need to be performed for all oLPNs planned on a pallet, the work creation level can be set to Pallet.
- A user exit is provided to allow customization of the work creation level as per specific requirements.
- Resource Group Eligibility - Resource Group Eligibility defines which resource groups, if any, are allowed to process a given work. In scenarios where resources are not modeled, resource groups are not required. However, when resources are modeled, the eligibility helps prioritize how work is executed. For example, consider a scenario where it is preferred to process orders using a unit sorter. If the sorter is at full capacity, the second preference is a putwall. In this case, the resource group eligibility would list sorter as the primary resource followed by putwall. This prioritization ensures that work is routed efficiently based on available capacity and operational preferences.

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- Inventory Check - Inventory Check is a critical validation process that ensures the required inventory is available before the work is released. It helps prevent creating tasks for allocation that cannot be executed due to inventory constraints. This is applicable for non-bulk flows.
- Resource Check - This determines whether resource capacities should be validated before assigning the work to a resource. While it is generally intuitive to perform resource checks when resource groups are defined, there are scenarios where skipping the check is appropriate. For example, if certain work (e.g., HOT work) must be processed immediately in a specific resource like sorter, regardless of its current capacity, assigning the work to a work release criterion with resource check disabled is appropriate. This ensures the priority work is not delayed due to capacity constraints.
- Open Work - This controls how long work remains open to receive additional allocations. It is useful in scenarios involving partial orders or chase allocations where inventory may arrive in stages. For example, if the building strategy is to create work at the oLPN level, the work can remain open until the oLPN is fully packed. Any subsequent allocations, such as chase, will be directed to the same resource handling the original work till the work is completed. However, keeping work open for extended periods can tie up resources longer, potentially impacting throughput.
- Priority and Cutoff Weight - This attribute determines how work is scored by controlling the relative importance of priority versus cutoff time. It accepts values between 0 and 1. For more details, refer to the Work Score documentation.
- Batch Together By - This determines how work is grouped for inclusion in the same Pack Wave or Resource Group Family batch. When set to "Order", all work (i.e., allocations grouped based on the work creation level) for the same order will be batched together. This ensures that all oLPNs associated with an order are processed and packed together, improving packing efficiency and order completeness.

Example: WT1, WT2, and WT3 are work release criteria assigned to the allocations during template determination. For every allocation, a work group key is generated. The work group key is a combination of the work creation level attribute value and work criteria. WT1 & WT2 have a work creation level of oLPN and WT3 has a work creation level of Order.

Order	oLPN	Allocation ID	Work Criteria	Work Group Key
O1	oLPN1	A1	WT1	oLPN1::WT1
O1	oLPN2	A2	WT2	oLPN2::WT2
O2	oLPN3	A3	WT3	O2::WT3
O2	oLPN4	A4	WT3	O2::WT3

Work creation will group the allocations based on the work group key and create work. If a work already exists with the same work group key, it will check for the open work status. If the status of work is less than the open work status, it will be added to the same work; otherwise, a new work will be created. For the above scenario, the output of work creation will be:

Order	oLPN	Allocation ID	Work Release Criteria	Work ID
O1	oLPN1	A1	WT1	WORK-1
O1	oLPN2	A2	WT2	WORK-2
O2	oLPN3	A3	WT3	WORK-3
O2	oLPN4	A4	WT3	

Work Release Criteria for Replenishment Resource Strategy

Replenishment work creation is also controlled via work release criteria, which are configured as a part of the replenishment resource strategy. Work release criteria attributes such as work creation level, open work and priority-cutoff score weight are not applicable for replenishment work. Replenishment work is always created at the allocation level.

- Resource Group Eligibility - Resource Group Eligibility defines which resource groups, if any, are allowed to process a given

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work. In scenarios where resources are not modeled, resource groups are not required. However, when resources are modeled, the eligibility helps prioritize how work is executed. For example, replenishment can be modeled by considering the capacity constraints of ASRS and the limited availability of induction stations, aiming to prioritize critical replenishment activities.

- Resource Check - This determines whether resource capacities should be validated before assigning the work to a resource. While it is generally intuitive to perform resource checks when resource groups are defined, there are scenarios where skipping the check is appropriate. For example, work is ideally processed in the decanting station, the station currently lacks additional capacity. If the business requirement is to ensure that replenishments proceed regardless of capacity constraints, then it is advisable to assign allocation to a work release criterion with resource check disabled.
- The work group key will be the allocation key and the work release criteria.

Business Updates

- Work is created and stored in the "work" entity.
- The new work is in "READY_FOR_CM" status and the allocations added to the work are also moved to "READY_FOR_CM" status.
- Based on the underlying allocations, the work type can have the following values:
  - Order - Pick/Pack and Pack Allocation (if work creation level was set to order)
  - oLPN - Pick/Pack and Pack Allocation (if work creation level was set to oLPN)
  - Allocation - Pick/Pack and Pack Allocation (if work creation level was set to allocation)
  - Pallet - Pick/Pack Allocations (if work creation level was to pallet)
- Pick work is created for Single_Pick and Pick allocations at allocation level for internal purposes for bulk allocation.
- For order and oLPN work types, the work size is fixed as 1. For allocation-level work, the allocation quantity determines the work size.
- Work group key is a combination of Order Number/oLPN Number/Allocation Key and Work Criteria Name. Work group key is not relevant for pick work.

Score Work

Work Score is a numerical value based on attributes of work. The higher the score, the greater the importance of the work.

The capacity manager algorithm uses a scoring logic to guide resource allocation decisions. Each work is assigned a score that serves as a factor for its relative importance. When evaluating multiple works, the algorithm compares scores to prioritize which work should have resources allocated.

The scores of the work records depend on factors such as:

1. relative priority of the work
2. relative cutoff for work
3. weight of priority vs cutoff as defined in the work criteria
4. Priority as Hot

Additional Notes on Scoring

- Work associated with a hot order receives a higher score compared to lower-priority work. This ensures that HOT work is clearly distinguished and prioritized by the capacity manager. For example, all other factors being equal, a low-priority work will have a lower score than one with a higher priority. HOT work is always given the highest priority.

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- Priority and Cutoff Weight - The work score is calculated as a weighted average of the priority and cutoff. For each work, the priority score and the cutoff score are computed internally.

For example, if the priority score is 100 and the cutoff score is 200, and the weightage is 80% for priority and 20% for cutoff, the work score will be:
Work Score = (0.8 * 100) + (0.2 * 200) = 120
In the reverse scenario, where the weightage is 20% for priority and 80% for cutoff, the work score will be:
Work Score = (0.2 * 100) + (0.8 * 200) = 180

- The weighted approach ensures that the final score reflects the relative importance of priority and cutoff in the decision-making process.
- If the cutoff time exceeds 48 hours from the current time, the score for the work is defaulted to 1. This applies regardless of the weightage configured for priority and cutoff time, ensuring that long-lead-time work is de-prioritized in scoring.
- The replenishment work must be scored when it is modeled for capacity planning. The score for replenishment work is determined from the score of the associated pack or pick/pack work based on dynamic sorting and assignment. This ensures alignment between replenishment priorities and downstream fulfillment activities.

Configuration

The work release criteria defined within the sort pack resource strategy incorporate the weighted values for priority and cutoff. These weightages are used to calculate the overall work score.

Business Update

The work score is persisted in the corresponding allocation in the attribute "*workscore*". No status transition happens for allocation and work.

Capacity Manager Model

The capacity manager model is the core decision-making engine, when invoked, it reviews the list of pending work and their associated allocations, selecting those that maximizes the total score. This selection is based on factors such as inventory availability and, where applicable, resource group capacities. Essentially, the model answers the following questions:

1. Does the work have sufficient inventory to be executed?
2. If yes, does the resource group allowed for this work have the capacity to execute this work?
3. If yes, and (if performing Resource Group Family batching) does a work resource have the physical capacity to execute this work?
4. If yes, is this the best work to select to satisfy the overall objective?
5. If yes, then assign the work to a resource group, and to a resource if performing Resource Group Family batching.

The capacity manager model evaluates inventory levels, resource group constraints, and available resource group capacity to identify and assign the most suitable work. In scenarios where resources are unavailable, the model defaults to checking inventory alone and releases only work that is not blocked by inventory constraints.

Override Constraints for Pick/Pack and Replenishment Allocations

The inventory check ensures that the inventory is available for an allocation to be processed further. The inventory check can be overridden to enable the release of allocations that are in "*Ready for CM*" status due to inventory unavailability. Rules can be configured to bypass the inventory validation. For configuration details, refer to the Work Release Configuration Guide.

This feature is helpful in cases where the cutoff time is nearing, clearing of the allocation pool at the end of a shift, or for aging

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allocations.

For example, if inventory is unavailable for aging allocations, an override rule can be configured using the attributes from the Work Release Allocation entity in the *Override Constraints* section as follows: "allocation.CreatedTimeStamp.hoursToCurrent < 2". This configuration skips the inventory check and forces the release of the allocations 2 hours post their creation.

> **Note:** Override Inventory Constraint should be configured only if the Inventory Check parameter is enabled in the Work Release Template.

Post Processing

Post the model run, a set of work records is assigned to a resource group (and to a work resource when performing Resource Group Family batching), and released for tasking.

Business Updates

Work

- Pick/Pack Work for non-bulk and Replenishment Work selected by the capacity manager moves to the status of "READY_FOR_TASKING".
- Pack Work selected by the capacity manager for bulk moves to the status of "READY_FOR_DISTRIBUTION".
- Pick Work selected by the capacity manager for bulk moves to the status of "READY_FOR_TASKING".
- Replenishment work and allocations belonging to work criteria, with no resource check and no resource group, and belonging to a reserve location, do not go through the capacity manager and go straight to "READY_FOR_TASKING".
- Resource group is updated to resource group assigned by capacity manager.
- Resource is updated to resource assigned by capacity manager (if assign resource = true for resource group, or if Resource Group is assigned to a Resource Group Family).
- Batch ID is updated to work and allocation by capacity manager in Pack Wave flow.
- Work not selected by work release remains in the status "READY_FOR_CM".

Allocation

- Allocations with the status "READY_FOR_CM", corresponding to PICK/PACK and Replenishment Work selected by capacity manager moves to the status of "READY_FOR_TASKING".
- Allocations with the status "READY_FOR_CM", corresponding to PACK Work selected by capacity manager moves to the status of "READY_FOR_DISTRIBUTION".
- Allocations with the status "READY_FOR_CM", corresponding to PICK Work selected by capacity manager moves to the status of "READY_FOR_TASKING".
- Allocations with the status "READY_FOR_CM", not selected by capacity manager remains in the status "READY_FOR_CM".

Adaptive Capacity

The Capacity Manager is responsible for releasing work based on the effective available capacity of the system. While physical resource capacities, such as unit sorters, putwalls and pack stations, are important, the most critical factor is the Task Manager buffer ratio. Each equipment has its own staging capacity, and work release must account for delays introduced by downstream processes like picking. To ensure consistent utilization of work resources, buffer capacity must be carefully maintained.

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While allowing users to define a buffer ratio is a good starting point, the system should also support dynamic, automated adjustments. These adjustments can be based on real-time data such as picking, sorting and packing throughput, as well as current workload in progress.

To arrive at a fair approximation of the adjusted value system requires the following statistics:

1. Average number of work in the system for the resource group
2. Average number of work in the packing area, i.e., pick completed
3. Average number of work in tasking or picking area
4. Picking Throughput - The average picking rate for the work resource group in terms of # of work
5. Equipment Throughput - The average packing rate for the resource group in terms of # of work per unit time
6. Prioritized Equipment Throughput - The average packing rate for the resource group in terms of # of work per unit time
7. System throughput - Throughput across all resource groups
8. The average cycle time of the work completed by the system: The time from Held to Completed
9. The average cycle time of the work by priority completed by the system: The time from Held to Completed

The accuracy of adaptive buffer adjustments improves with the depth and quality of historical data available. Given the sensitivity of auto-adjusting buffer ratios, the system adopts a cautious and controlled approach to upward and downward changes. Rather than reacting to short-term fluctuations, the system monitors for consistent trend indicating either underutilization or overloading of resources. Adjustments are made only when consistent patterns are observed, and always within pre-defined, configurable bounds to ensure operational stability.
The statistics listed above can be captured at a global level (i.e., one per work release cycle) or for each resource group.

Configuration

The adaptive statistics parameters can be configured at one of the two levels:

1. Work Release Parameters: Applies across all work resource groups
2. Work Resource Group: Allows fine-grained control for individual resource groups

If statistics needs to be tracked at the resource group level, users can define the Adaptive Capacity Parameters, which control the automatic adjustment of buffer capacities. Adaptive behavior for each resource group can be toggled using the adaptiveEnabled flag. If this flag is set to true, but no adaptive parameters are defined for the work resource group, the system will default to the parameters defined at the work release parameters level. The hierarchical configuration ensures flexibility while maintaining a fallback mechanism for consistent behavior.

1. Stats Calculation Window - This defines how far back (in time) statistics need to be evaluated for adjusting the buffer ratio
2. Min time to wait before adjustment - This defines the mandatory lag between consecutive buffer ratio adjustments
3. Time to Confirm trend - This defines the time to wait to confirm a trend (increase/decrease Buffer ratio) before making any adjustments
4. Allowed Deviation - This defines the allowed deviations which the system can ignore while establishing a trend for starvation/overflow of capacity
5. Capacity Ratio Change By - This defines the buffer ratio adjustment (increment/decrement) to make when adjustment is needed

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Business Updates

Buffer Ratio for eligible work resource groups is updated when all necessary conditions are met.

Warehouse Resources

Work release enables users to create a logical representation of physical resources unit sorters, putwalls, and pack stations to model equipment capacities, staging areas, resource groupings, operational constraints and other relevant characteristics. These resources are used in the warehouses to sort and/or pack inventory.

Glossary

- **Work Resources:** This defines the physical equipment, like sorter chutes, putwall cubbies, and pack stations.
- **Resource Group:** This entity logically groups a set of similar work resources. Each resource group needs to be tied to a resource group type.
- **Resource Group Type:** This defines the type of resource group. A resource group type can be sorter, putwall, or a pack station.
- **Resource Group Family:** This is a collection of resource groups that is used for grouping work for a given unit sorter into a Pack Wave.

When allocations are created and imported into Work Release, the system first determines the appropriate templates. Based on the configured work creation level, work is created and scored to establish the priority. Once work is created and prioritized, the system evaluates both resource capacity and resource eligibility to ensure that tasks are executed efficiently and only when sufficient capacity is available.

To perform this evaluation, work release follows a structured decision-making process. This process begins by checking whether the required resource capacity exists. If capacity is available, the system proceeds to assign the appropriate resources, such as resource groups or families, before releasing tasks for execution. This approach ensures that work is only assigned when operational conditions are optimal, helping maintain throughput and task integrity.

These assignments are based on the type of work to be done and ensure that the right set of resources is aligned with the work requirements. Once resources are assigned, the system performs a capacity check to confirm that the selected resources can handle the workload. This is followed by a verification of minimum criteria, which ensures that the basic conditions for tasking such as batch size, throughput expectations, or location constraints are met. If both the capacity and criteria checks are successful, the system proceeds to select allocations that fit within the available resource capacity. This includes consideration of any buffer capacity, which provides flexibility in planning and helps prevent overloading. These allocations are then marked as "READY_FOR_TASKING", indicating that they are prepared for execution and can be picked up by the task manager.

The final step in the flow is to perform task criteria checks, which validate that all operational requirements such as timing, labor activity, and resource availability, are satisfied before the work is released. This ensures that tasks are executed under optimal conditions, maintaining both task integrity and resource throughput. This logical flow supports adaptive and efficient task management by aligning work with available resources, minimizing delays, and maximizing operational performance.

Buffer Capacity

In operations where multiple resources are involved in completing a task, such as picking followed by packing, or picking, putwall sorting, and then packing, it is essential to maintain a continuous flow of work across stages. This is where buffer capacity plays a critical role. Buffer capacity allows the system to release more work than the immediate physical capacity of downstream resources. For example, pickers need to stay ahead of packers or putwall sorters to prevent idle time and ensure uninterrupted throughput. If picking were to begin only after a pack station becomes available or a putwall cubby is cleared, it would create delays

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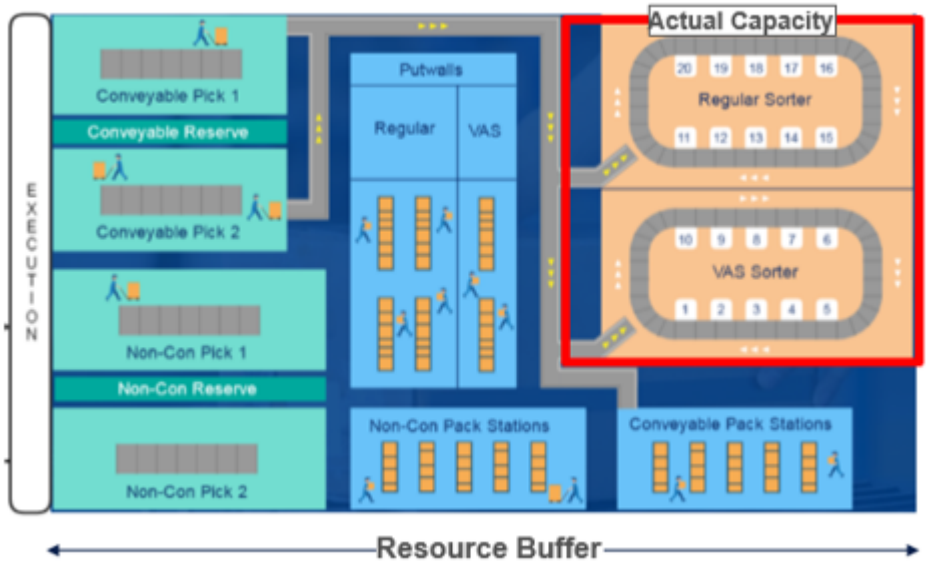
and reduce efficiency. To avoid such bottlenecks, buffer capacity enables the system to release more work than the actual resource capacity. This means that even if a downstream resource is temporarily full, upstream tasks like picking can continue, with the assumption that the downstream resource will catch up as work progresses. This approach helps maintain momentum and ensures that each stage of the workflow remains productive.

Consider the following example:



Each individual work resource within a resource group has a defined capacity. This could be based on throughput, batch size, or the number of items it can handle within a given time frame.
The total capacity of a resource group is calculated as the aggregate of the capacities of its individual work resources. For example, if a sorter resource group includes 3 chutes, and each chute can process 100 units per hour, the total capacity of the group is 300 units per hour. This aggregation allows the system to plan work more effectively by understanding the full potential of the grouped resources.

The primary objective of work release is to ensure that the most critical or high-priority work is released first. This prioritization is based on a scoring mechanism that evaluates work against business rules, urgency, and operational constraints. By releasing only the most important work, the system ensures maximum utilization of available resources, avoids bottlenecks, and maintains throughput efficiency.

Let us take an example of a sorter resource group which has 3 Chutes (Work Resources).

Resource Group	Resource Group Type	Work Resource	Base Capacity
Sorter A	Unit Sorter	CH01	10
		CH02	10
		CH03	10

"Sorter A" has a base capacity of 30 units, it can handle 30 units of work concurrently. However, releasing work strictly up to this base capacity can lead to inefficiencies. When work is released to match the exact capacity, resources may remain idle while waiting for work to arrive. Similarly, once the resource is occupied, picking operations may pause because the system detects that the downstream resource is full. This creates a bottleneck and disrupts the flow of operations. To overcome this, work release uses buffer ratios to release more work than the base capacity, ensuring a continuous flow and minimizing idle time. There are two types

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of buffer ratios:

- Capacity Manager Buffer Ratio: This ratio is used to determine how much work can be assigned to a resource group beyond its base capacity. A higher buffer ratio gives the system more flexibility to plan additional work, creating a larger pool of work for task creation. This helps achieve higher pick density, meaning more efficient picking with fewer interruptions. The capacity manager buffer ratio is not used for Pack Wave batching.

  Available Capacity for Resource Group Assignment = Sum of base capacity of Resources * (1 + Capacity Manager Buffer Ratio) - WIP

  Base Capacity of Sorter A = 30 units
  Capacity Manager Buffer Ratio = 3
  Work in Progress (WIP) = 10 units
  Available Capacity for Resource Group Assignment = 30 * (1 + 3) - 10 = 120–10 = 110
  This means the system can assign up to 110 additional units to the sorter work group.

- Task Manager Buffer Ratio: This ratio is used during the execution phase and represents the actual buffer available for releasing work to the resource. It controls how much work is released to maximize resource utilization without overwhelming the system. The task manager buffer ratio is not used for Pack Wave batching.

  Available Capacity at Task Creation = Sum of base capacity of Resources * (1 + Task Release Manager Buffer Ratio) - WIP

  Base Capacity of Sorter A = 30 units
  Task Manager Buffer Ratio = 2
  Work in Progress (WIP) = 10 units
  Available Capacity at Task Creation = 30 * (1 + 2) - 10 = 90–10 = 80

  Out of the 110 units made available, the Task Release Manager uses this pool to generate tasks. However, only 80 of these units, selected by the Capacity Manager, are actually released for execution.

Resource Group Capacity Threshold

The Work Resource Group Capacity Threshold controls when new work is released to a work resource group based on its available capacity. It ensures that the work is released only when a minimum percentage of capacity is available, helping prevent the creation of low-density tasks. For example, imagine a sorting/packing area where multiple work compete for limited resources. If work is released whenever capacity becomes available, resources may be overloaded with small tasks, causing inefficiency.
Consider the following scenario:

Work Resource Group Total (Maximum) Capacity	Work Resource Group Capacity Threshold	Available Work Resource Group Capacity
20	0.7 (70%)	14

If the Work Resource Group Capacity Threshold is set to 0.7, work is released only when at least 70% of the resource group capacity is available. In this case, work is released only when there is an available capacity of 14.

Resource Group Family/Pack Wave

Resource Group Family refers to either a single unit sorter or a logical grouping of resources and resource groups. The resource group family determines the level at which Pack Waves are generated and released. WM provides a functionality to divide a picking wave into smaller batches. Batches refer to a grouped set of work that are processed together as a unit. These batches are typically created to optimize how the work is released, routed and executed across available resources. A batch can include a collection of items, orders, or containers that share similar processing requirements. The "Batch Together By" configuration in the Work Release

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Criteria determines how work is grouped into batches during the pack wave or resource group family batching. This ensures that all work sharing the same value for the selected attribute are included in the same batch. For example, if "Batch Together By" is set to Order, all work (i.e., allocations created at the same work creation level) associated with a specific order will be grouped into a single batch. This enables all oLPNs related to that order to be packed together, supporting efficient and consolidated packing operations.

By grouping the work, work release can better manage capacity constraints, reduce idle time, and ensure smoother flow through the operation. For example, if the chute has a limited capacity in terms of weight or volume, forming a batch ensures that the total assigned work stays within those limits.

Work release provides the ability to create and release batches in a coordinated manner. When a unit sorter operates in batch mode, requiring a fixed amount of work to be released and inducted at once, resource group family batching can be enabled using the configuration associated with resource group. This configuration defines how many batches should be released simultaneously and sets a minimum threshold that must be met before a batch can be created. Each resource within a resource group family can be configured with specific capacity limits. A unit sorter is typically configured to handle only one batch at a time.

When work is assigned to a resource group that belongs to such a family, the Capacity Manager evaluates whether the available work meets the configured minimum threshold. If it does, one or more batches are created. Work is then assigned to the most suitable resource (for example, a chute) within the highest-priority eligible resource group. This assignment considers the size of each work record relative to the configured capacity of the resource. The outcome is a set of well-structured batches that are ready for processing, aligned with the operational constraints.

For example, consider a unit sorter configured as a resource group family with two resource groups, each containing 100 chutes (200 chutes in total). If each chute can hold one work record and the minimum batch threshold is set to 80% of total capacity, then each batch must contain at least 160 work records. Given 600 available work records, the Capacity Manager will create approximately three batches of 200 work records each, assuming each record fits within the chute's capacity. Any remaining work records will be held until enough are available to meet the minimum threshold for the next batch.

Manual Batch Release and Batch List Screen

When the `ManualBatchRelease` flag is enabled, the system requires manual intervention to release batches for tasking. In this mode, once batches are created, they are not automatically released. Instead, they remain in a "HELD_FOR_TASKING" status. This allows users to review, prioritize, or coordinate the release of work based on operational needs before any tasks are generated.

To proceed with releasing these batches, users must navigate to the Batch List UI. From there, they can select one or more batches that are currently in the "Held for Tasking" status and initiate the **Release** action. Once released, the system transitions the associated allocations to "READY_FOR_TASKING" status, and the corresponding tasks are created. This manual control ensures that only approved batches move forward in the workflow, offering flexibility in managing workload distribution.

After a batch is released, it transitions to "Released" status. At this point, users can click the **Refresh** button in the Batch List UI to update key summary fields such as Total Orders and Total oLPNs. These fields provide visibility into the contents of the batch and help confirm that the correct work has been released.

In addition to manual release, the system provides a **Release for Induction/Put** action. This feature is used when batches reach the "Picking Completed" status. Selecting this option raises a Material Handling Equipment (MHE) event, signaling that the inventory is ready for induction. The system then communicates with the appropriate equipment based on the event configuration. For more details on the structure and content of these events, refer to the MHE Event List section.

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Finally, the system also supports a **Reset** action for batches that are still in the "HELD_FOR_TASKING" status. When this action is triggered, all associated allocations and work records are reset to "READY_FOR_CM" status, and the batch itself is moved to "Cancelled" status. Additionally, any distribution entries related to bulk allocations are removed. This provides a way to cleanly undo batch creation if the work needs to be reassigned or re-evaluated.

Task Release Manager

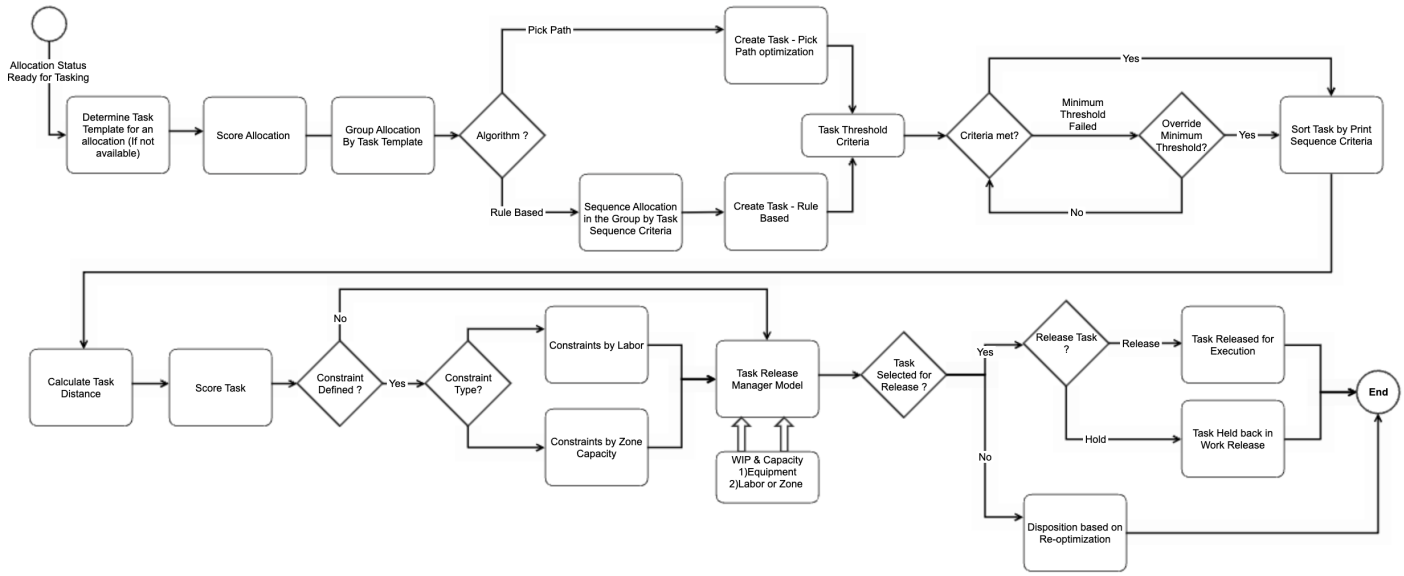
The task release engine is responsible for creating and releasing tasks based on allocations selected by the capacity manager. It uses user-defined rules for allocation selection, sequencing and threshold criteria to generate tasks. Allocations can be grouped either to minimize the travel distance across tasks or according to a user-defined sequence. Once tasks are created, they are evaluated against optional constraints such as labor availability, zone capacity limits, and equipment resource constraints. Tasks that meet these criteria are then released for execution. The configuration of these constraints is optional and can be tailored to operational needs.

The capacity manager identifies the high-priority work(s) and assigns appropriate resource groups. The task release manager then creates tasks from these work records, applying the defined rules and evaluating them against any configured constraints.

Task Release Manager
Score Allocations → Create Task → Calculate Task Distance→ Score Task→ Task Release Manager Model→ Release Task

The task release manager performs the following functions:

- Creates tasks for allocations marked with status "READY_FOR_TASKING". Task creation can be rule-based or driven by pick path optimization.
- Computes the travel distance for tasks using DC layout.
- Assesses tasks against labor availability, zone capacity, and equipment resource capacity before releasing them for execution. This is not applicable for for Pack Wave batching flow.
- Persists the selected tasks in Pick Pack component and relevant messages are sent to the external systems.



Score Allocation

Tasks are created by grouping allocations. Before task creation, all the allocations marked with status “READY_FOR_TASKING” or “READY_FOR_DISTRIBUTION” are scored for two key purposes:

1. Task Creation: Allocation scores help prioritize which allocations should be included in tasks, with preference given to high-scored allocations.
2. Task Score: The scores of individual allocations are used to calculate the overall score of the resulting task.

This scoring process ensures that the task creation aligns with operational priorities.

Configuration

For Pick/Pack and Pack allocation, the priority and cutoff score will be calculated based on the priority and cutoff weight defined in the work release criteria.

Pick/Pack, Pack and Pick Allocation Scoring

The scoring for these allocations is dependent on the following parameters:

1. Priority
2. Cutoff
3. Allocation marked as HOT.
4. Time-based Score Adjustment: If an allocation has been aging and remains in a particular status beyond a configured duration (defined by the "*bumpUpScoreTimeLimit*" parameter in Work Release Parameter), its score is increased to reflect the aging. This adjustment helps prioritize older allocations over new ones, ensuring timely processing.
   For example, suppose the bump up is set to 10 minutes. An allocation associated with a work that has already been released will have a higher score than one that has not yet been released. Similarly, an allocation that has been waiting on a resource for 20 minutes will be scored higher than one that has only waited for 5 minutes. This ensures that aging allocations gain priority over time, helping prevent delays and bottlenecks in task execution. The bump up time reflects the time limit after which the work is assumed to age and is given a bump up in its score to give it an edge over other allocations.
5. The pick allocation score is determined by doing a weighted average score of pack allocations.

Replenishment Allocation Scoring

As a part of the capacity manager run, the replenishment allocations are scored to establish the relative importance of replenishments.

Business Updates

The allocation score is updated using the allocation's "score" attribute.

Create Task

The allocations with status "READY_FOR_TASKING" are eligible for task creation. Tasks are generated based on rules defined in the task criteria, which specify how tasks should be created and include additional attributes related to labor task execution. Each task is assigned a task ID, the format of which can be configured using the Next Up configuration for "TaskId".

Task Priority is calculated by scaling the task score to a range between 0 and 49 using the following formula:

Scaled Score = Task Score * (49.0 / 150.0)

Task Priority = 99 - Scaled Score

If the resulting priority is less than zero, a default priority of 1 is assigned. The final priority value ranges from 1 to 99. The task score itself is a weighted sum of the average score, maximum score, and bump-up score. Additionally, if the isHotOrder flag is set to True, a bonus score is added further to elevate the task's priority.

Task creation can either be rule-based or driven by pick path optimization.

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Rule-Based Task Creation

Rule-based Task Creation refers to the process of creating tasks based on a pre-defined set of business or operational rules. These rules are typically configured to align task creation with specific priorities, constraints or workflows.

Allocations that match the same task criteria are grouped together for task creation. This includes allocations in the ”READY_FOR_TASKING” status.

The parameter “Should task creation consider allocations that are not yet ready for tasking?“ defines how allocations are evaluated during task creation. It determines whether all allocations within a defined group, such as those sharing the same pallet, must be ready before a task is created. When enabled, the system waits until every allocation in the group is in ”READY_FOR_TASKING” status. If even one allocation is held due to inventory or resource constraints, none of the allocations in that group will be considered for task creation. For example, if the operations want to group tasks by pallet and avoid fragmented release until all allocations part of the pallet are ready, they can configure a distinct value task threshold as *allocation.plannedParentContainerId - Minimum Value=0 and Maximum Value=1.* If there are 2 allocations for the same pallet, one is ready to be tasked and the other is in “READY_FOR_CM” status due to inventory constraint check, both the allocations are held until they are both ready to be tasked.

Once grouped, allocations associated with a specific task template are optionally sorted using the task sequencing criteria defined in the task criteria.

A new task is then created, and the sorted allocations are added to the task sequentially. As each allocation is added, the system validates whether the task exceeds the configured maximum threshold criteria. If it does, the task release logic uses a look-ahead counter configured in the task criteria and skips the current allocation and evaluates the next one, continuing until the look-ahead counter limit is reached. If look-ahead is not required, the counter should be set to zero. If look-ahead is not allowed, the current task is sealed, and a new task is created from the next eligible allocation.

After all allocations are added, the task is evaluated against the minimum threshold criteria. A task is created only if it meets or exceeds all minimum thresholds and does not violate any maximum thresholds, otherwise it is discarded. However, override rules can be set up to bypass the minimum thresholds and allow task creation when required. If print sequencing is configured, the task details are re-sequenced to align with the defined print order.

Pick Path Optimization Algorithm

Pick path optimization is a meta-heuristic (MHS) algorithm used for advanced task creation. Unlike the rule-based method, this approach creates tasks by minimizing the travel distance. It leverages the DC Layout component to determine the shortest path between locations, considering constraints such as directionality and vehicle types. The primary objective is to optimize task grouping in a way that reduces the overall travel time and increases operational efficiency. While tasks are created based on the layout-driven logic, they still adhere to task break rules, minimum threshold criteria, and batching configurations defined in the task criteria. Because the algorithm inherently sequences tasks based on spatial proximity, explicit task sequencing or print sequencing rules are not required. Optimized task creation is recommended for facilities handling high-volume orders, where reducing picker travel time can significantly enhance throughput. It is particularly beneficial in environments with complex layouts, such as multiple zones, directional aisles, or vehicle-specific paths, where it contributes to improved operational efficiency and productivity.

Travel Distance Calculation using Layout information

Warehouses can customize picking routes by specifying specific start and end locations for tasks. When configured, the algorithm incorporates these travel attributes into the travel distance calculation during task creation. If no stand and end locations are defined, the algorithm defaults to using the allocation's pick location as both the start and end location. For configuration

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details, refer to the Location Configuration Guide.

Task Threshold Criteria

Task Threshold criteria define the maximum and minimum limits that must be met for a task to be considered valid.

- **Maximum Task Thresholds:** The maximum thresholds act as break criteria. They determine the capacity at which a task should be closed and a new task created. For example, if a template specifies a maximum weight of 25 lbs per task, the system checks whether adding a new allocation would exceed this limit. If it does, the current task is sealed, and a new task is created. Otherwise, the allocation is added to the existing task, provided no other maximum thresholds are violated.
- **Minimum Task Thresholds:** The minimum thresholds serve as acceptance criteria. They ensure that any generated task meets a minimum limit enforced on an attribute. Any task created must exceed the minimum value defined for the corresponding attribute. For example, a user may configure a minimum of 10 allocations per task to ensure operational efficiency.

In summary, maximum thresholds define when a task should break, while minimum thresholds ensure that a task is substantial enough to proceed. For a task to be valid, it should either meet or exceed all the minimum thresholds and should not exceed any of the maximum thresholds.

The task threshold criteria support four types of task threshold criteria:

- **Distinct Value Limits:** This rule category defines how tasks are segmented during task generation based on the number of distinct values allowed for specific attributes. The user specifies both the attribute name and a limit value. For example, if the user selects Aisle Number as the attribute and sets the limit to 1, the task created will have AIDs (Allocation IDs) that are located in the same aisle. If the limit is set to 2, the task may include AIDs from up to two distinct aisles. Attributes used in distinct value limit rules can be alphanumeric. Both minimum and maximum limits can be configured to control task grouping behavior.
- **Capacity Limits:** This rule category defines how tasks are segmented during task creation based on the total value of a specified numeric attribute. The user configures the rule by selecting an attribute (e.g., Quantity) and setting a limit value. For example, if the user selects Quantity as the attribute and sets the limit to 50, the task created will have a total quantity that does not exceed 50 units across all AIDs (Allocation IDs). Capacity limit rules apply only to numeric attributes.
  Both minimum and maximum thresholds can be configured.
- **Range Limits:** This rule category controls how tasks are split during task generation based on the range, the difference between the maximum and minimum values of the specified numeric attribute. The user defines an attribute (e.g., Priority) and a range limit. For example, if Priority is selected with a limit of 5, the task created will have all AIDs (Allocation IDs) grouped into a single task have priority values within a 5-unit range. If the difference between the highest and lowest priority exceeds this limit, a new task is created. Range limit rules apply only to numeric attributes. Unlike other threshold rules, range limits are used exclusively for task breaking and do not support minimum threshold enforcement.
- **Feasibility:** The feasibility limit type allows the use of mathematical or logical expressions as attributes when configuring threshold criteria. These expressions enable advanced validation rules during task creation. For example, you can define an expression to ensure that all the items within a task share the same country of origin. To implement this, import the required expression into work release, set the limit type to Feasibility and assign the attribute to the label of the imported expression. This provides flexibility in enforcing complex business rules that go beyond simple attribute checks.

Note: To use the feasibility limit type, please contact your Manhattan representative.

Override Minimum Threshold

The minimum task thresholds configured in the task threshold criteria ensure that the minimum limit is met or exceeded for a task to be created. These minimum task thresholds can be overridden to force the release of allocations in "Ready for Tasking" status when the minimum threshold is not met. Override rules can be configured to override the minimum task thresholds. Force release of

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such allocations can be useful when the cutoff time is nearing, allocations are aging, etc. The override threshold should be configured only if minimum task thresholds are set up. The override rules can be configured using the attributes from Work Release Allocation and oLPN entities.

For example, consider the Task Threshold configuration, which specifies a minimum of 3 oLPNs per task.

Task Threshold Configuration

Limit Type	Attribute	Minimum	Maximum
Distinct Value	oLPN	3	8

The override rule configured is "allocation.CreatedTimeStamp.hoursToCurrent < 2". If the allocations only amount to 2 oLPNs and if at least one of the allocations satisfies the override rule, then a task is created with 2 oLPNs.

Split Allocation

When a single recall or replenishment allocation exceeds the configured task maximum threshold, the system can be configured to split the allocation into multiple smaller allocations based on the maximum capacity task thresholds or UOM. For configuration details, see the Work Release Configuration Guide.

If the task already has some allocations and the subsequent allocation added is over the threshold, then the allocation is not split and is added to the new task created.

When the parameter is set to *Break by Maximum Capacity Threshold*, the allocations are split based on the maximum capacity value defined in the task threshold configuration. Allocation Quantity, Weight, and Volume attributes in the Capacity limit task threshold are supported.

Example: Allocation UOM is UNIT, and the capacity limit type is configured with item volume

Configuration:

Task Release Template	Limit Type	Attribute	Maximum Value	Split Allocation?
Template1	Capacity	Volume	15	Break by maximum capacity threshold

Item's Unit Volume = 1

Original Allocation:

Allocation	Task Release Template	Status	Allocation Type	Allocation UOM	Allocation UOM Quantity	Quantity	Weight	Volume
AID02	Template1	Ready for Tasking	Replenishment	UNIT	0	20	20	20 (20 * Item's Unit Volume (1))

The solution splits the original allocation up to the maximum capacity configured (that is, 15) for the template. A new allocation is created with 15 units, and the original allocation is retained at 5 units.

Allocation	Task Release Template	Allocation Type	Allocation UOM	Allocation UOM Quantity	Quantity	Weight	Volume
AID02	Post for template assignment	Replenishment	UNIT	0	~~20~~ **20 - 15 = 5**	~~20~~ **20 - 15 = 5**	~~20~~ **20 - 15 = 5**
AID02-01	Post for template assignment	Replenishment	UNIT	0	15	15	15

When the parameter is set to *Break by UOM*, the allocations are split into a single shippable UOM. For example, if the allocation has 5 packs/25 units and the pack quantity is 5, it is split into 5 allocations, each containing 1 pack of 5 units.

Notes:

- No further split happens regardless of the allocation exceeding the maximum capacity threshold if the allocation UOM is LPN.
- UOMs such as Units, Bundles, Packs, and Subpacks are eligible for split based on a maximum capacity threshold or UOM.
- During the split allocation process, the allocations are not split to contain partial UOMs.

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Task Distance Calculation

Once tasks are created, the system calculates the total distance required to complete each task. If no start or end location is specified, this calculation assumes that the location of the first allocation is the starting point and the location of the last allocation is the endpoint. Task details are sent to the DC layout component, which returns the total travel distance for the sequence of locations involved in the task. Additionally, it provides the distance between each consecutive pick location within the task.
For example, consider a task **T1** with three allocations: **A1**, **A2**, and **A3**, located at **L1**, **L2**, and **L3**, respectively. If each location is 5 meters apart:

- The **total task distance** for T1 is **10 meters** (L1 L2 L3).
- The **distance to next pick** is:
  - A1 A2: 5 meters
  - A2 A3: 5 meters
  - A3: 0 meters (final pick)

This distance data helps optimize task execution.

Configuration

DC Layout component should be enabled and available for distance calculation.

Business Update

For the task selected by the task release manager, the total distance is updated on "travelDistance" on Task. On task detail, the distance to the next pick is stored in "distanceToNextPick".

Score Task

The objective of the task manager within the work release process is to prioritize and release the most important tasks, subject to labor constraints. Each task is assigned a score that reflects its relative importance, calculated using weighted values derived from its constituent allocations.

The task score is determined based on the following parameters:

- Average allocation score across all allocations in the task
- Maximum allocation score within the task
- Bump Up score applied to aging allocations
- Presence of allocations belonging to Hot work within the task

These task scores are then used by the task release manager model to decide which tasks should be released for execution, ensuring that high-priority work is addressed efficiently.

Configuration

Work Release Parameter has the configuration of Score Weight for Average Allocation Score, Score Weight for Maximum Allocation Score, Score Weight for Bump up Score and Bump up Score time limit. These will be used as weights for the calculation of the task score.

Task Release Manager Model

The Task Release Manager (TRM) is a mathematical model responsible for determining which tasks should be released for execution. It evaluates tasks based on their scores and available capacity constraints. Two types of tasks generated are:

- Replenishment Tasks

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- Picking Tasks

The Task Release Manager engine, when invoked, reviews the list of tasks and their associated allocations. It selects tasks that maximize the overall score while considering constraints such as:

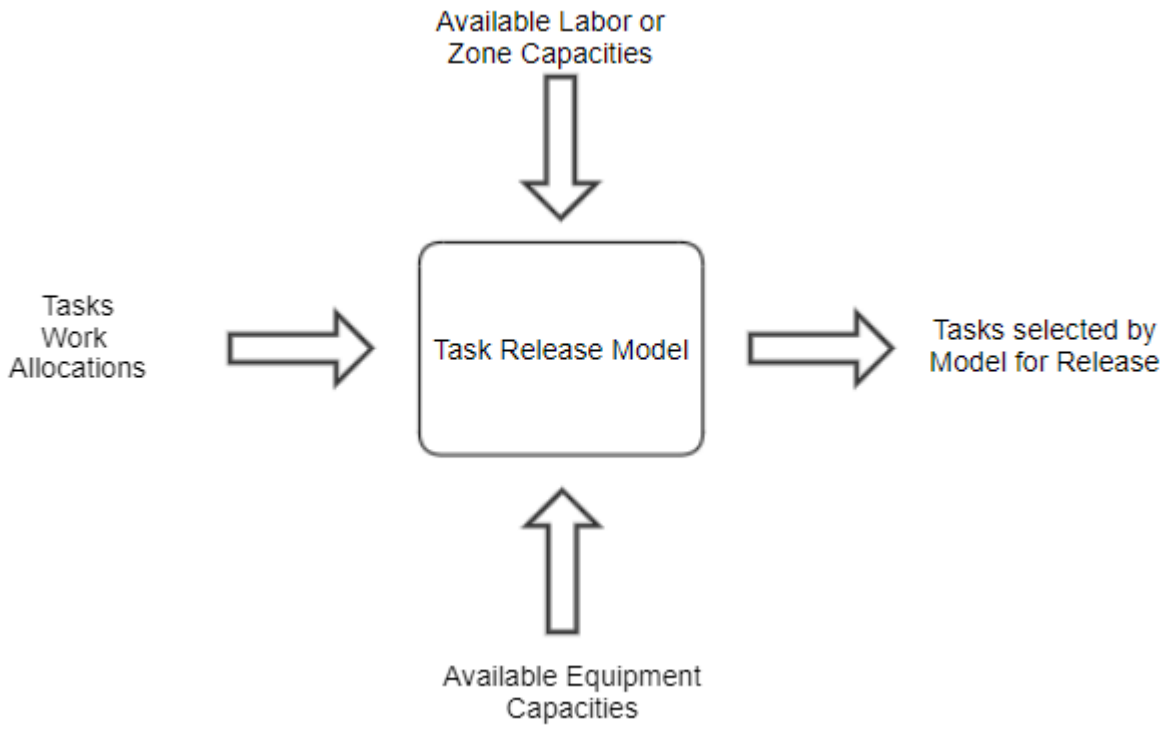
- Labor availability
- Zone picking capacity
- Resource group capacities (if applicable)

The model answers the following key questions:

1. Does the resource group have enough capacity to execute the work in this task? The resource capacity in this case is based on task release manager buffer ratio.
2. Is there sufficient labor or zone capacity to execute the task? This is evaluated using the released and WIP tasks to determine the available capacity.
3. Assign task group to tasks: This is applicable for non-zone picking tasks. The task release manager determines the list of eligible tasks groups for a task based on task type, INT, work group and work area of the first task detail.

Task release is balanced across eligible task groups to optimize capacity utilization.
In the absence of labor, zone or resource capacity constraints, the task release manager releases all created tasks. If tasks are batched, task release manager ensures that either all tasks in a batch are released together or none are released.

Note: This model is not used for Pack Wave release process.



Task Release Constraints

The task release constraints help determine which tasks can be released based on operational limitations. The task release

constraints can be of two types:

- **Labor Constraints** - Tasks are released based on the availability of pickers or a pre-defined task count limit. For example, the system may be configured to release a maximum of 10 tasks.
- **Zone Capacity Constraints** - Tasks are released based on the available capacity within specific picking zones. This is particularly relevant in zone-based picking operations, where each zone has its own throughput limits.

Labor Constraint

In this operational scenario, tasks are released based on either the availability of pickers or a pre-defined task count limit. Each task is evaluated for eligibility against task groups responsible for pulling inventory from the source location. Intermediate task groups, which handle subsequent legs of the task, are not modeled for labor constraints. The tasks are assessed against:

- Labor capacity as defined within the task group
- Resource capacity including sorter, putwall and pack station availability

The goal is to release the maximum number of tasks without violating any of the configured constraints.

Note: These constraints are not considered for the Pack Wave release process.

Task Group

Task group configuration enables control over the picking operation by setting constraints based on the number of available pickers within each task group. When configured, the work release process honors these constraints and releases tasks accordingly. Additionally, task groups can be configured with duration-based capacity constraints, which are useful in scenarios where tasks need to be released to occupy available pickers for a specific time period. For example, the system can release tasks that are expected to keep pickers engaged for the next hour. These constraints are stored in the Task Group entity.

During each work release run, the system queries the Task component to retrieve the Active User Count, which reflects the number of users currently clocked in for each task group. Labor constraints can either be user-defined or systematically determined based on real-time labor availability and operational parameters.

To calculate the active user count, the system looks at users who are currently active (enabled and belonging to the organization, facility, and profile) and eligible to be assigned to a task group. Then, for each user, the system tries to find their assigned task group. If the user has a current task group already set, that task group is used. If not, the system uses the task group the user is eligible for. In cases where the default task group behavior in User Defaults is not set, system uses the eligible task group. Finally, the users are grouped by their task groups and counts how many unique users are in each task group. This helps give an accurate picture of how many active users are available in each task group.

Zone Capacity Constraint

Zone Picking

The zone picking flow enables modeling and constraining picking operations based on the number of containers present in a zone at any given time. In this approach, items are divided across multiple picking zones, and pickers are pre-assigned to specific zones. Pickers identify their zones by scanning location barcodes or entering their picker ID. Unlike traditional labor modeling based on task count, zone picking models labor constraints using the number of containers (e.g., totes or oLPN boxes) actively worked on within a zone. When an order spans multiple zones, containers move through a conveying system to subsequent zones for continued picking. In each zone, the assigned picker scans the container and picks the required items. Once all picks in a zone are completed, the container is routed to the next zone or sent to a packing or sortation station.

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The objective of work release in this context is to identify and release oLPNs that maximize zone capacity utilization while minimizing shortages, overages, and congestion. Zone capacity is modeled based on the maximum number of containers that can be staged in a zone at any time.

- For example, **Zone 1**, being a large zone, may support up to **100 containers**.
- In contrast, **Zone 2** may only accommodate **50 containers** due to space or operational constraints.

Notes: These constraints are not considered for the Pack Wave release process.

Static zone picking can be enabled in the Allocation Strategy. In zone picking, each allocation includes a Zone ID, which is stored in the field "PIKR_NBR". When allocations are imported, the associated zone information is captured in the "locationGroup" attribute. This allows the system to correctly associate each allocation with its designated picking zone.
Assume that the zone configuration entity is configured as shown below. The zone capacity is in terms of oLPN/Containers.

Zone	Sequence	Zone Capacity	Zone Capacity Buffer Ratio	Entry Scan ID	Exit Scan ID
ZN1	1	5	20%		
ZN2	2	5	20%		
ZN3	3	5	20%		
ZN4	4	5	20%		
ZN5	5	5	20%		
ZN6	6	5	20%		

When a task is created for zone picking, its details are organized in the sequence defined by the Zone configuration entity. For each task, the system determines a task path, which represents the sequence of zones the task will traverse. For example, consider task T1 that involves picking items from the zones, Zone 1, Zone 2, Zone 4 and Zone 6. The system will generate the task path ZN1 ZN2 ZN4 ZN6. This path ensures that the picking sequence aligns with the configured zone order, optimizing task execution and minimizing travel time.

Understanding Task Path and Zone Period Capacity

The task path defines both the sequence of zones a task will traverse and the timing associated with each zone. From a planning perspective, each zone in the task path is treated as a distinct time period. For example, consider a task that moves through the following zones:

- Zone 1 during Period 1
- Zone 2 during Period 2
- Zone 5 during Period 3

In this scenario, the task consumes 1 unit of capacity in each zone during its respective time period.

- In Period 1, capacity is consumed from Zone 1.
- In Period 2, capacity is consumed from Zone 2.
- In Period 3, capacity is consumed from Zone 5.

Now, consider another task T2 with the following zone sequence ZN3 ZN4 ZN5. The corresponding zone period capacity matrix would represent the task's presence and capacity consumption across time periods and zones.

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Period/Zone	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
Period 1	T1		T2			
Period 2		T1		T2		
Period 3				T1	T2	
Period 4						T1

In scenarios where an order involves picking from all available zones, each zone is mapped to a distinct time period. For example, if an order spans four zones, it is assumed, purely from a planning perspective, that the task will occupy four corresponding time periods. The zone period capacity matrix represents the demand placed on each zone's capacity by newly planned work. Regardless of the number of picks within a zone, a task is assumed to occupy only one time period per zone. As the task progresses, the work release process receives execution feedback. This feedback enables more accurate planning by refining estimates of:

- Work-in-progress (WIP)
- Available capacity per zone and time period

This dynamic adjustment helps align planning assumptions with real-world execution, improving overall system responsiveness and resource utilization.
For tasks that have already been released, the system calculates a zone period capacity matrix for zones that are yet to be picked. For example, consider a task T3 that was previously with the task path ZN1 ZN2 ZN3. At the time of release, its zone period capacity matrix would have represented the expected capacity consumption across the remaining zones. This matrix helps in planning and managing zone-level workloads before actual execution begins.

Period/Zone	Zone 1	Zone 2	Zone 3
Period 1	T3		
Period 2		T3	
Period 3			T3

Execution feedback indicates to work release that task T3 has completed picking in Zone 1. As a result, when work release runs next, the zone period matrix will be recalculated to reflect the remaining zones.

Period/Zone	Zone 2	Zone 3
Period 1	T3	
Period 2		T3

Notice that Period 1 is now Zone 2 as compared to Zone 1 earlier. The current zone of a task is determined using the task zone history entity. The system identifies the zone associated with the most recent activity, based on the following criteria:

- Latest pick completion message
- Latest entry timestamp
- Latest exit timestamp

The zone corresponding to the most recent of these is considered the task's current location. For example, assume zones are configured to track both entry and exit scans.

- At time T0, work release receives an entry scan for Zone 1.
- Between T1.....Tn, pick completion messages are received for items in Zone 1.
- However, no exit scan has been received yet.

In this scenario, for planning purposes, Work Release assumes that the task is still occupying capacity in Zone 1, since there is no signal indicating that the task has exited the zone.

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Calculating Available Zone Capacity

To assign new work efficiently, the system calculates available capacity for each zone and time period by subtracting the currently occupied capacity (work-in-progress) from the total zone capacity, including buffer.

Zone Capacity = (1 + Buffer Ratio) * Zone Capacity

Available Zone Capacity = Zone Capacity - Work In Progress

For example, consider Zone 1 has base capacity of 5 with a buffer ratio of 20%.

Zone Capacity = (1 + 0.2) * 5 = 6

Now, assume 2 containers are currently present in Zone 1 during Period 1, and they have not yet exited the zone.

Available Zone Capacity = 6 - 2 = 4

The available capacity is used to determine whether new tasks can be assigned to Zone 1 during Period 1.

The model uses the calculated values for demand on capacity, occupied capacity and available capacity to determine which tasks are eligible for release.

Using Feedback to determine Task Location and Zone Capacity

Execution feedback plays a critical role in identifying the current zone of a task, which directly impacts the calculation of available capacity in each zone. To free up capacity, the system must determine whether a task has exited its current zone. This determination is based on picking feedback:

- If all picks for a zone are completed, Work Release assumes the oLPN has moved out of that zone, making its capacity available for new tasks.

In automated systems, where oLPNs are moved via conveyors, Work Release can use divert information to more accurately track oLPN movement. These systems rely on photo scans to detect and divert oLPNs to the appropriate zones. The movement of oLPNs across zones is recorded in the Task Zone History entity. In some scenarios, photo scan data may be shared with Active WM, providing Work Release with real-time oLPN location updates. This location data is essential for accurately calculating zone capacity and managing task flow. The Task Zone History entity serves as the central record of a task's progression through zones, enabling precise planning and capacity tracking.

Zone Recirculation

In certain execution scenarios, a task may not follow its originally planned zone path. This behavior is common in implementations where containers or totes can be recirculated, and the downstream execution system opportunistically modifies the path based on real-time conditions. To determine the updated zone path, Work Release evaluates open allocations for zones that precede the current zone. If any picks in those earlier zones remain incomplete, those zones are appended to the end of the task path, following the sequence defined in the Zone configuration. For example, the initial task path determined by Work Release is ZN1 ZN2 ZN4 ZN6. The task is released, and feedback is recorded in the "taskZoneHistory" entity. Based on feedback, work release identifies the current zone as ZN4. Picks in ZN1 are completed, at least one pick in ZN2 remains incomplete. In this case, work release assumes that ZN2 was skipped and appends it to the end of the path. The updated task path becomes ZN4 ZN6 ZN2. This dynamic adjustment ensures that all required picks are completed, even if the original zone sequence is altered during execution.

Pack Wave Task Release

When tasks are created for a pack wave batch (grouped by resource group family), they are released together in a single batch. Each resource group family is configured with:

- A maximum number of batches to release
- A maximum number of batches to hold
- A batch size threshold

At any given time, the system maintains a set of in-progress batches on the floor and a set of held batches ready for release. As batches are completed, new batches can be released. If capacity allows, additional batches may also be created to maintain the

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configured maximum.

The work release process evaluates:

- The current count of held and in-progress batches
- The effective open capacity for additional batches
- The available batches for the resource group family

It then selects the best batch(es) to release based on the average score of the work within each batch. These selected batches are released as described in the next section.

A batch size threshold refers to the minimum size of a batch that allowable to be released, expressed as a percentage of the active resource capacity. It is the lowest acceptable number of work records that should be available before a batch can be released. This is configured as a value between 0 and 1.

For example, if the active resource capacity is 200 chutes and the batch size threshold is set to 0.8, then the minimum batch size is 80% of 200 which is 160. A batch cannot be released unless it contains at least 160 work records.

Controlling Task Creation Across Batches

The `IgnoreBatchesDuringTaskCreation` parameter determines whether task creation can consider allocations from different batches.

- By default, one task is created per batch.
- If the parameter is enabled, the system ignores batch IDs on allocations and groups all allocations with the same task release template into a single task.

This allows pack wave and non-pack wave allocations to be combined into a single executable task, improving flexibility and resource utilization.

Release Tasks

Tasks are evaluated for release using either the Task Release Manager Model or the Pack Wave Release logic, depending on the configuration and context. Only tasks that meet the criteria defined by these models are selected for release.

Based on the "Should the system release task?" parameter maintained in the task release template, the task will be released for execution. If the parameter is disabled, tasks are created in “Held” status in Work Release.

Business Updates

When tasks are evaluated for release, they fall into two categories: selected or non-selected. Each of these follows a distinct handling process.

Non-Selected Tasks

If the task is not selected for release, it is discarded. The associated allocations are handled based on the re-optimization level set in the work release parameter.

Selected Tasks

If a task is selected for release, it is stored in the "task" entity with details recorded in "taskDetail". Allocation status is updated based on the type of work:

- **Non-Pack Wave Work**
  - If "Should the system release task?" is enabled
    - Allocations are moved to "`RELEASED`".

    - If the work was in "`READY_FOR_TASKING`", it is also moved to "`RELEASED`".
    - Otherwise, the work status is updated to

- **Pack Wave Work**
  - Allocations for released batches are moved to "`RELEASED`".
  - Allocations for held batches remain in "`READY_FOR_TASKING`".
- Execution Feedback
  - Once the task is released, update messages are sent to external systems.
  - As execution progresses, the Feedback Manager receives updates.
  - These updates are used by Work Release for:
    - WIP calculation
    - Adaptive logic
  - Refer to the **Feedback Manager** documentation for more details.

Pick Cart Flow

A pick cart is a mobile unit equipped with multiple slots or compartments. Each slot is designated for a specific picking task, which could be:

1. A single oLPN - Items that are picked go directly into a container that gets shipped.
2. An order - Items are grouped based on customer order requirements.
3. Batch of picks across multiple oLPNs - Items picked for multiple orders are grouped by logical units, e.g., for a zone.

Each task corresponds to a distinct destination or grouping, picked into an oLPN directly, or reusable totes, or cart slots. Pick Carts can be used for both cubes and non-cubed flows.

oLPN Pick Cart

In pick into oLPN scenario, inventory is picked directly into the shipping container and work release process facilitates the creation of optimized pick carts. These carts are designed to streamline picking operations by aligning with either rule-based logic or pick path optimization. When pick carts are created, work release aims to achieve the following:

- **Fitment:** Ensure each oLPN fits within the physical constraints of the cart.
- **Capacity Utilization:** Maximize the use of available cart space for container assignment.
- **Travel Efficiency:** Minimize the total distance traveled across all pick tasks (applicable only when using pick path optimization).

In the Pick to oLPN process, oLPNs are selected based on predefined criteria and sequenced using configured attributes. These oLPNs are then grouped into pick carts according to a threshold that determines how many containers can be assigned to each cart. To ensure organized and efficient picking, each oLPN is assigned to a specific slot or sub-slot within the cart.

Cart capacity is configured to ensure that each container fits within its designated slot and that the cart can accommodate the total number of containers assigned. The constraints such as slot dimensions and overall cart layout are defined to reflect the physical structure of the cart and prevent overloading.

While task thresholds determine the number of containers per cart, the Pick Cart Plan ID governs the exact placement of each oLPN. This configuration mirrors the cart’s physical layout, specifying the number of shelves and the dimensions of each shelf (width,

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depth, and height). Each shelf is treated as a slot, and slots may be further divided into sub-slots to optimize space usage. Slot and sub-slot IDs can be barcoded to support confirmation during picking. Sub-slots inherit the height and depth of their parent slot, while their width is divided proportionally.

The pick cart plan ID configuration allows for capturing the layout of the pick cart. Each shelf is treated as a slot, and every slot is assigned a unique Slot ID, which can be barcoded if slot-level confirmation is required during downstream picking. Slot dimensions are recorded to ensure that the assigned containers fit appropriately. To optimize space and organization, slots may be further divided into sub-slots. These compartments share the same height and depth as their parent slot, while the width is distributed proportionally based on the number of sub-slots.

For example, if a slot has a width of 9 units and is divided into 3 sub-slots, each sub-slot will have a width of 3 units. Sub-slots can also be assigned barcoded IDs to support sub-slot level pick confirmation.

Below is the representation of the physical pick cart viz-a-viz pick cart plan ID configuration:

For the above representation, below is the relationship between slot and sub-slot:

SLOT	SUB-SLOT
SLOT 1	SUB SLOT 1.1
	SUB SLOT 1.2
	SUB SLOT 1.3
SLOT 2	SUB SLOT 2.1
	SUB SLOT 2.2
	SUB SLOT 2.3
SLOT 3	SUB SLOT 3.1
	SUB SLOT 3.2
	SUB SLOT 3.3

Process Flow

During task creation in the Pick to oLPN flow, the system evaluates each assigned oLPN to determine its critical dimensions. These dimensions are identified during the cubing process and are used to guide slot assignment within the pick cart. Since oLPNs can

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vary in size, the task creation algorithm ensures that each oLPN is placed in a slot that accommodates its dimensions while respecting the physical constraints of the pick cart. By default, the oLPN is oriented along the slot's width, provided its width and height do not exceed the slot's depth and height. If this orientation is not feasible, the algorithm attempts to place the oLPN along its length, ensuring that its width and height still fit within the cart's depth and height. This approach prioritizes optimal fit and efficient use of space. During slot assignment, the system follows a structured approach:

- **Step 1**: Identify slots with the maximum width and sort them in descending order.
- **Step 2**: Identify slots with the maximum height and sort them in descending order.
- **Step 3**: If multiple slots have the same width and height, they are sorted alphabetically or numerically based on slot names.

When pick path optimization is enabled, the algorithm further enhances efficiency by minimizing overall travel distance across pick tasks. It evaluates available oLPNs and assigns them to pick carts in a way that balances space utilization with travel optimization.

This method ensures that slot selection is both dimensionally appropriate and operationally efficient.

Key Considerations

Each pick cart task corresponds to a single cart. The task includes a set of oLPNs that are assigned to specific slots or sub-slots within the cart based on configuration rules.

- **Slot and Sub Slot Assignment**
  - Each oLPN is assigned to one slot only and will not span multiple slots.
  - A slot may contain multiple oLPNs, depending on capacity.
  - Sub-slots, which are subdivisions of slots, can also hold multiple oLPNs.
  - If an oLPN spans multiple sub-slots, it is assigned starting from the first sub-slot in its sequence.
  - Sub-slots are allocated in the order they are defined in the configuration.
- **Confirmation Logic**
  - The current Pick Cart process supports slot level confirmation.
  - If a slot contains more than one oLPN, individual oLPN confirmation is required to ensure accuracy during picking.
- **Pick Cart Identification**
  - Each cart is assigned a unique pick cart number, generated based on the configuration for "`PickCartId`" under the Work Release "Next Up Number".
  - This number is stored in the "pickCartNumber" attribute of the pick cart task.
- **Monitoring Progress**
  - Users can view the status and contents of each pick cart task through the Pick Pack Tasks UI, which provides visibility into task execution and cart assembly.

Tote Pick Cart (Cart of Totes)

In pick into tote scenario, inventory is picked into reusable totes that are small container or bins rather than into the shipping container. This is commonly used in warehouses that require high-speed, high-accuracy picking. Each tote is typically assigned to a specific order or group of items. Tote templates are used to group picking allocations into totes, which serve as reusable picking containers. Depending on the configuration, a tote can be assigned:

- All picks for a single oLPN
- All picks for a single order

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- Picks for multiple oLPNs that will be sorted and consolidated in a later process

This approach supports both cubed and non-cubed flows. In non-cubed scenarios, picks for the entire order are placed into a tote. Users then create and assign oLPNs during the packing process. Once totes are filled, they can be grouped into pick carts for further processing.

Tote templates are especially effective when using standard-sized containers, as they simplify cart assembly and improve picking efficiency. They also support batch picking for non-cubed orders, allowing multiple orders to be picked together and sorted later. This configuration enables flexible, scalable picking operations while maintaining accuracy and control over container usage.

The pick into tote flow is designed to support the following key considerations:

- Each pick cart has a fixed number of slots. Each slot is assumed to contain one sub-slot, meaning only one tote can be assigned per slot.
- The system assumes that all slots have infinite capacity. In other words, the algorithm does not validate whether a tote fits in a slot, it assumes that any tote type can be accommodated.
- Each tote is associated with a Tote Type, which defines its physical characteristics such as length, width, height, and weight limits. These attributes can be configured to support operational constraints.
- To enable pick cart tote task creation, users must select "Pick into Cart" as the task execution mode. Refer to Tote Template configuration for more details.
- Tasks are created at the cart level, and task details are grouped into totes as part of the algorithm.
  - Cart level break by (e.g., *PickZoneId*, *Resource Group Family*, *Resource Group*, *Pack Wave Batch*) continues to be supported.
  - Additionally, tote level break by are supported (e.g., *Resource Group Family*, *Resource*, *Pack Wave*), allowing for more granular control.
- The configuration supports including complete work records, oLPNs, or orders within a tote, depending on operational requirements.

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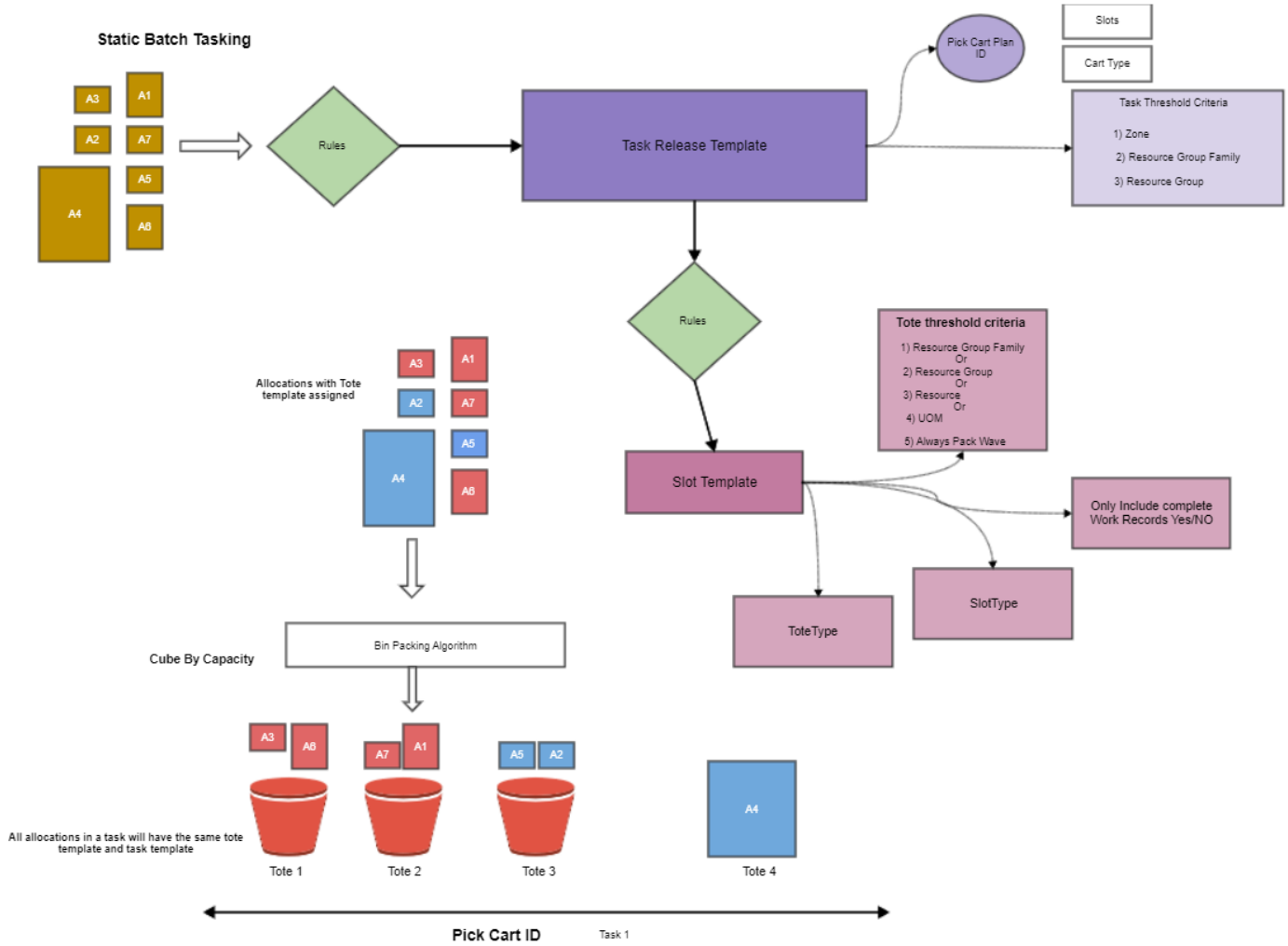

Hybrid Pick Carts

- Tote Pick Cart - Carts that can accommodate only totes.
- Hybrid Pick Cart - Carts that can accommodate both totes and oLPNs. (Algorithm during optimization decides the ratio of totes and oLPNs for a given task)

- Tote Type - Defines the dimensions of the tote. (Big, Medium, Small)
- Slot Template - Defines the rules that govern the allocations that can be grouped together for a slot via tote or oLPN.

Work Release supports advanced planning of pick carts that include both totes and oLPNs. This capability enables the system to generate picking tasks that are optimized across multiple dimensions, including:

- **Pick path efficiency**: Minimizing the total distance traveled during picking.
- **Tote utilization**: Ensuring optimal use of tote capacity.
- **Order priority**: Honoring the urgency or service level of orders.
- **Destination-specific requirements**: Adapting to the unique handling needs of different fulfillment destinations.

In many warehouse environments, a single pick cart may carry totes destined for different processing areas. For example:

- Some totes may be routed to a unit sorter, which automatically sorts items by order or destination.
- Others may be sent to a putwall, where items are manually sorted into cubbies for packing.

Each destination may impose different requirements on the totes:

- Unit sorter totes are often smaller and may not require complete work records (i.e., they can be partially filled or contain items from multiple tasks).
- Putwall totes, on the other hand, may require complete task fulfillment within a single tote to streamline manual sorting.

To manage these variations, tote-specific requirements are modeled using slot templates in the system. Each slot template defines the physical and logical characteristics of a slot on the pick cart. When the slot is intended to hold a tote, the slot type is set to 'Tote'. This configuration allows the system to:

- Assign totes to appropriate slots based on their destination and handling rules.
- Ensure that the pick cart layout supports both operational efficiency and downstream processing needs.
- Maintain flexibility in task creation while still enforcing destination-specific constraints.

By leveraging work release's ability to plan pick carts with both totes and oLPNs, operations can achieve a high degree of picking efficiency while accommodating diverse fulfillment workflows. The use of slot templates with tote-specific configurations ensures that each tote is handled according to its destination's requirements, whether it's a unit sorter, put wall, or another processing area.

Bulk Allocation

Bulk allocations refer to a type of allocation used in warehouse operations where picking and packing tasks are not immediately tied to specific orders or containers such as oLPNs. Instead of creating a one-to-one relationship between pick allocations and pack allocations at the time of inventory pull, bulk allocations allow for a more flexible grouping. In this model, a group of pick allocations for a particular item is associated with a group of pack allocations, but the exact pairing between them is deferred.

This approach is particularly useful in high-volume environments where inventory is pulled in bulk and later distributed to orders based on demand or packing logic. The key aspect of bulk allocation is that the inventory being picked is not pre-assigned to specific orders or containers. Instead, the work release process determines how the picked inventory is matched to packing tasks. This allows for dynamic decision-making based on real-time conditions such as order priority, chute availability, or packing station capacity.

By decoupling the pick and pack stages, bulk allocations provide operational flexibility and can help optimize resource utilization. However, they also require careful coordination during work release to ensure that inventory is correctly assigned and that packing tasks are generated in alignment with the order requirements.

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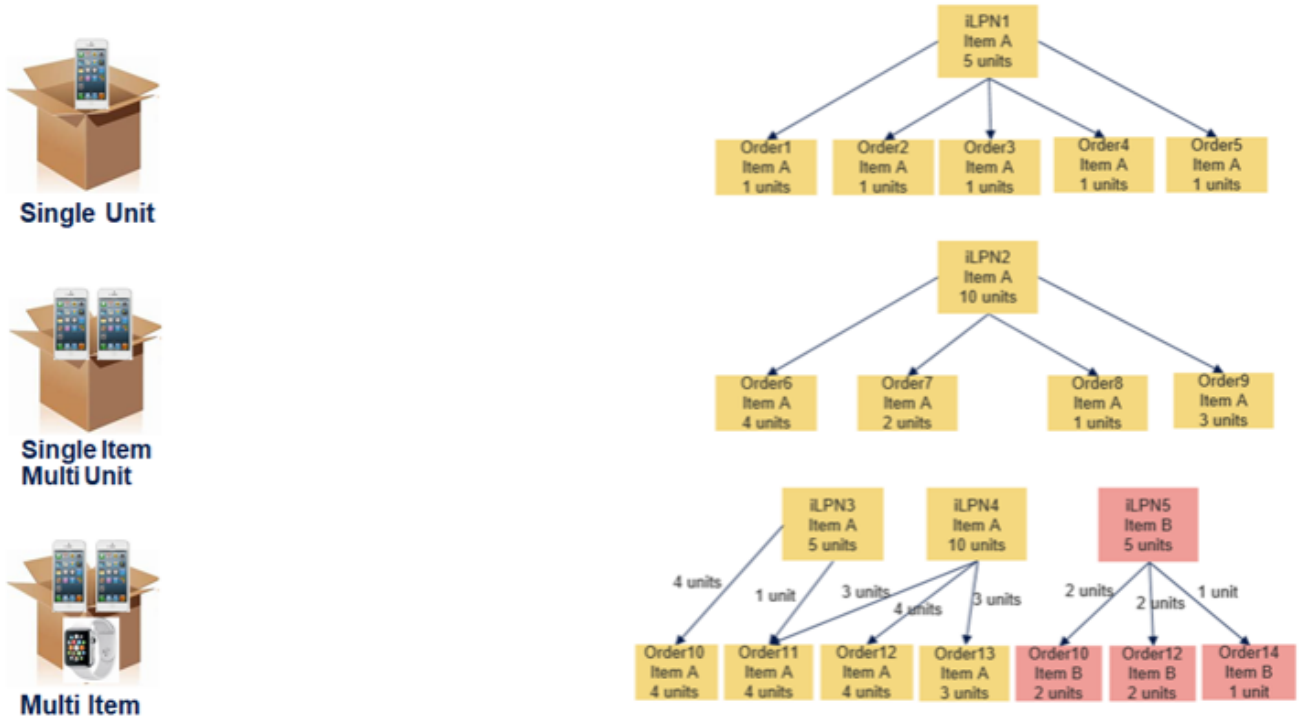

Bulk Allocation Method

The Bulk Allocation Method defines how inventory is allocated to orders during bulk picking and packing operations. It distinguishes between two types of bulk allocation methods:

- Singles Bulk
- Bulk

These methods determine how inventory, typically represented by iLPNs, is distributed across orders and how orders are fulfilled from available inventory.

In the **Singles Bulk** method, each iLPN is allocated to multiple orders, but each individual order is fulfilled entirely by a single iLPN. This ensures that no order is split across multiple iLPNs. This method is commonly used for items that are either single-unit or multi-unit but of a single item type, where simplicity and packing efficiency are prioritized. The key benefit of this approach is that it guarantees order integrity, each order is packed from one iLPN, making downstream processes like packing and shipping more straightforward.

In contrast, the **Bulk** method allows for more flexibility. Here, iLPNs can be allocated to multiple orders, and orders themselves can be fulfilled using inventory from multiple iLPNs. This means that an order may be split across several iLPNs depending on availability and capacity. While this introduces complexity in packing, it enables better utilization of inventory and supports high-volume operations. To manage this complexity, the work release process aims to minimize the number of splits per order, ensuring that fulfillment remains as consolidated as possible.

The choice between Singles Bulk and Bulk allocation is made at the time of allocation and is based on operational requirements, item characteristics, and fulfillment strategies. Understanding and configuring the appropriate bulk allocation method is essential for optimizing inventory flow, reducing packing errors, and improving overall efficiency in bulk processing environments.

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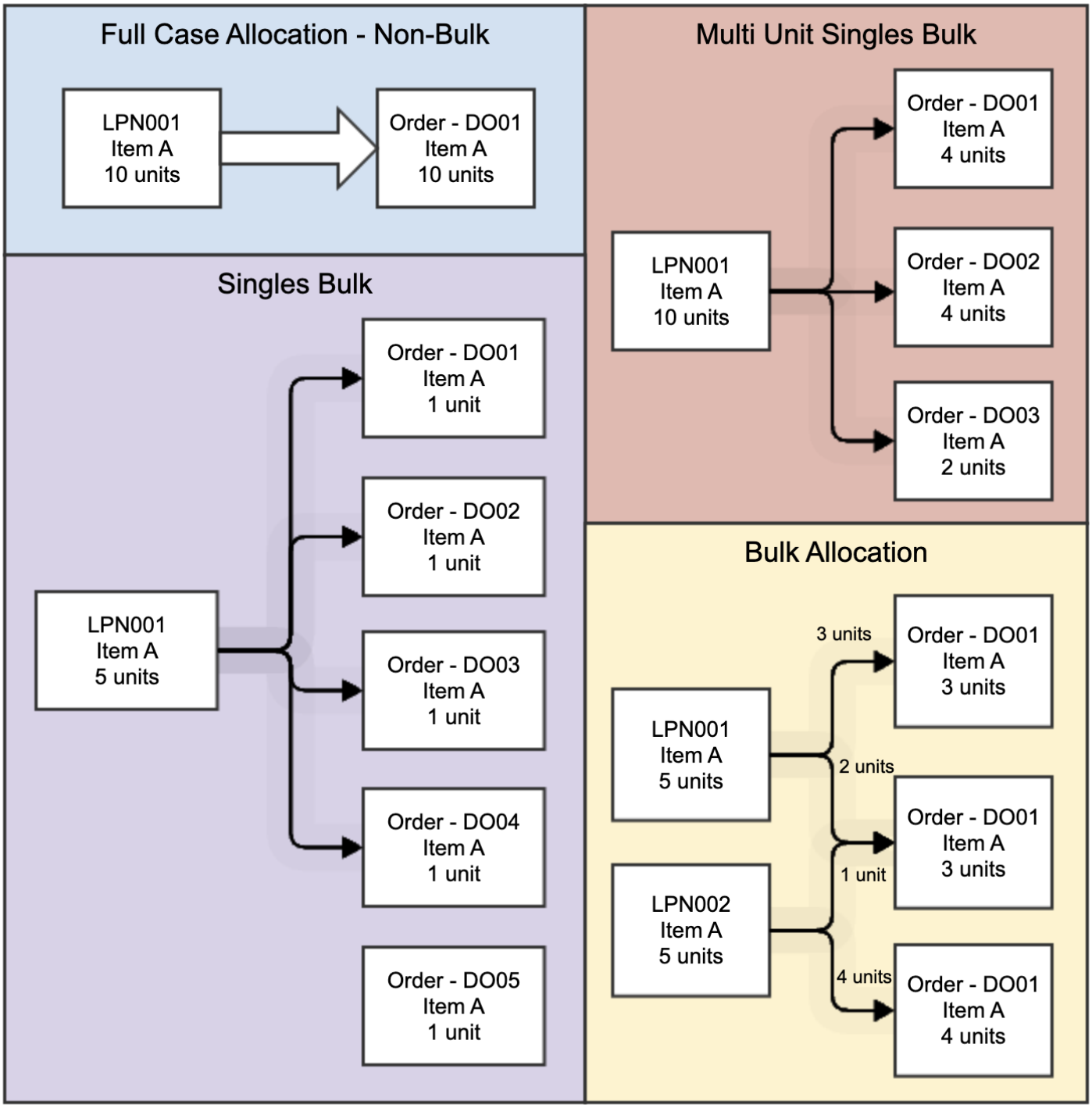

Bulk Group ID

During the allocation process, the system typically groups order needs that share the same inventory criteria. This helps streamline picking and packing by consolidating similar inventory requirements. However, in certain operational scenarios, it may be necessary to restrict this behavior to meet specific business rules. For example, based on order criteria, operations may need to route orders from one set of stores to a particular location, while orders from another set of stores are sent elsewhere, even though both sets share the same inventory criteria. To support this kind of controlled separation, the system uses the concept of a Bulk Group ID.

The Bulk Group ID acts as a logical boundary that defines which pick and pack allocations can be matched during work release. By assigning a Bulk Group ID to a set of allocations, the system ensures that only allocations within the same group, and using the

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same bulk allocation method, are considered for matching. This is particularly useful when operational rules require that the fulfillment of orders be contained within a specific context, such as a wave or a store group.

Another common use case for Bulk Group ID is to restrict the matching of pick and pack allocations to within a single wave. In this scenario, the Bulk Group ID is set at the wave level, ensuring that allocations from different waves are not mixed during work release. This helps maintain wave integrity and simplifies downstream processes like packing. If no such restriction is needed, the Bulk Group ID can be omitted. In that case, pick and pack allocations will not carry a Bulk Group ID, and the system will allow broader matching across allocations that share inventory criteria.

Ultimately, the Bulk Group ID, combined with the bulk allocation method, defines the scope within which pick and pack allocations are matched. This mechanism provides flexibility and control, allowing operations to enforce grouping rules that align with business needs. For more details on how to configure and apply Bulk Group IDs, refer to the Allocation Process documentation.

Refer to the example below for the explanation of these concepts. Work Release matches the pick and pack allocations only when they share the same Bulk Group ID and use the same bulk allocation method. In the illustration, this is represented by matching colors.

AID	Type	Item	Qty	LPN	Group ID	Bulk Allocation Method
A1	PICK	Item A	2	LPN01		Singles
A2	PICK	Item A	2	LPN03	G1	Singles
A3	PICK	Item A	5	LPN05	G2	Singles
A4	PICK	Item A	3	LPN07	G1	Bulk
A5	PICK	Item B	3	LPN08	G1	Bulk
A6	PICK	Item B	3	LPN09	G1	Bulk

AID	Type	Item	Qty	Group ID	Bulk Allocation Method
A11	PACK	Item A	1		Singles
A12	PACK	Item A	1		Singles
A13	PACK	Item A	1	G1	Singles
A14	PACK	Item A	1	G1	Singles
A15	PACK	Item A	2	G2	Singles
A16	PACK	Item A	3	G2	Singles
A17	PACK	Item A	2	G1	Bulk
A18	PACK	Item A	1	G1	Bulk
A19	PACK	Item B	2	G1	Bulk
A20	PACK	Item B	2	G1	Bulk
A21	PACK	Item B	2	G1	Bulk

Capacity Manager Processing for Bulk

In bulk allocation scenarios, the Capacity Manager plays an expanded role beyond just assigning resources. In addition to determining which resources such as chutes or pack stations should handle the work, it also manages the distribution of pick allocations to pack allocations. This distribution process ensures that inventory pulled during picking is correctly matched to packing tasks, based on operational constraints and real-time system status. The system considers the bulk allocation method, bulk group ID, item and item attributes during the distribution of pick allocations to pack allocations.

When performing this distribution, the Capacity Manager considers several factors:

- What work has already been released for tasking
- What has been released to the floor
- What inventory has already arrived at the resource groups

It also performs an inventory check to validate the availability of items for pick allocations. The results of this distribution process are stored in an entity called "`bulkDistribution`", which tracks how pick and pack allocations are linked.

The timing of this distribution can vary depending on the configuration. It may be performed during the work release process, or it

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can be delayed until the packing stage. If distribution is delayed until packing, work release does not generate any "WORK" records for the allocations. Instead, the pack allocations are marked as "Work Complete", and the pick allocations proceed through task creation.

This behavior is controlled by a parameter in the allocation strategy called “When should system distribute bulk allocations to the group?”. This configuration allows operations to choose whether bulk distribution should occur early during work release or later during packing, depending on their workflow needs. Choosing the right timing for bulk distribution is critical for balancing system performance, task accuracy, and operational flexibility.

Feedback Manager

In outbound warehouse operations, associates sort and pack customer orders using designated process areas such as cubbies or chutes. Once a task is completed, the space it occupied becomes available for new work. To keep Work Release informed of these changes, the Feedback Manager sends messages indicating updates in the equipment capacity. This creates a continuous feedback loop that helps Work Release maintain an accurate view of resource availability. Additionally, the Feedback Manager tracks task progress and completion, ensuring that Work Release has real-time visibility into execution status.
Feedback processing ensures that work release receives accurate and timely updates about the status of work being executed. These updates are used to:

- Reflect the current status of a specific work item
- Track how long a work item has remained open
- Release resources for new work once existing tasks are completed
- Measure the dwell time of a resource
- Support re-prioritization of allocations

Types of Feedback

- **Picking Feedback**
  - Provides updates on pick completion within zones.
  - Helps determine whether a task has exited a zone and whether capacity can be freed.
- **Work Feedback**
  - Communicates task-level progress and completion.
  - Typically originates from the resource where the work is executed.
  - Used to manage resource assignment and update task status.

Picking Feedback

After work is released, picking is the next step in the outbound fulfillment process. This step may be executed using tasks created during the release process or directly through picking allocations. To track progress and support planning, the picking process generates feedback events whenever a pick is completed. These events are critical for maintaining visibility into task execution and resource utilization. Work Release consumes these feedback events to:

- Update the status of allocations
- Reflect progress in the associated work
- If applicable, update the status of the related pack wave batch

This ensures that the system remains synchronized with real-time execution and can make informed decisions about releasing additional work.

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The following matrix summarizes the feedback touch points:

Flow	Transaction	Feedback Generation	Message Type & Content	Allocation Updates	Work Updates	Pack Wave Batch Updates
Batch Picking	Pick to Tote (RF,MHE) Pick Tote Cart	Feedback event generated on: Completion of each detail OR Cancellation of each detail	Feedback Type : NULL Task ID, Allocation key,Time Stamp,Task Group	Allocation status updated to "Picking Completed" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Picking is started and completed	Work status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the work are cancelled. "Work Completed" if no resources are assigned.	Batch status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.
Packing	Pack from Tote (RF, MHE)	Feedback event generated on: Completion of detail OR Cancellation of detail	Feedback Type " PACKING" Task ID, Allocation key, Time Stamp, Task Group	Allocation status updated to "WorkCompleted" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Cancelled" if all allocations for the work are cancelled. Work Complete will not happen here as it would be triggered by the Resource feedback in the form of Chute Close or Clear Cubby.	Batch status updated to "Cancelled" if all allocations for the batch are cancelled. Batch Complete will not happen here as it would be triggered by the Resource feedback in the form of Chute Close or Clear Cubby.

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Flow	Transaction	Feedback Generation	Message Type & Content	Allocation Updates	Work Updates	Pack Wave Batch Updates
Single Step Pick & Pack	Pack Cube Directed (RF, MHE) Pack oLPN Cart			Allocation status updated to "Work Completed" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the work are cancelled. "Work Completed" if no resources are assigned else triggered by the Resource feedback in the form of Chute Close or Clear Cubby	Batch status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.
Replenishment Pulling	RF Pull Case List	Feedback event generated on: Completion of detail OR Cancellation of detail		Allocation status updated to "Picking Completed" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the work are cancelled.	Batch status updated to "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.
Zone Picking	MHE Pack Pick Cart	Photo Eye Scan Entry	Feedback Type : NULL Task ID, Allocation key,Time Stamp,Task Group	Allocation status updated to "Picking Started" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Picking Started" if more open Allocation exists "Cancelled" if all allocations for the work are cancelled. "Work Completed" if no resources are assigned.	Batch status updated to "Picking Started" if more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.

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Flow	Transaction	Feedback Generation	Message Type & Content	Allocation Updates	Work Updates	Pack Wave Batch Updates
		Photo Eye Scan Exit or if Location Group is not configured for Exit Scan	Feedback Type : NULL Task ID, Allocation key,Time Stamp,Task Group	Allocation status updated to "Picking Completed" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the work are cancelled. "Work Completed" if no resources are assigned.	Batch status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.
	RF Pack Pick Cart - Tote Flow	On completion of Pack Pick Cart for the tote, feedback message is triggered		Allocation status updated to "Picking Completed" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed	Work status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the work are cancelled. "Work Completed" if no resources are assigned.	Batch status updated to "Picking Started" if more open Allocation exists "Picking Completed" if no more open Allocation exists "Cancelled" if all allocations for the batch are cancelled.
	RF Pack Cubed Directed / RF Pack Pick Cart - oLPN Flow	On completion of packing and when the oLPN goes to "Packed" status, feedback message is triggered		On successful completion of WORK, update Allocation status to "WORK COMPLETED" for all the allocations associated with the WORK.	Work status updated to "Work Completed" "Cancelled" if all allocations for the work are cancelled.	Batch status updated to "Work Completed" "Cancelled" if all allocations for the batch are cancelled.

Work Feedback

The order fulfillment process may separate picking and packing into distinct steps. This is common in flows that require additional sorting or packing after picking, depending on the characteristics of the order. To ensure warehouse throughput goals are met, one or more resources in the fulfillment process are designated as critical resources. These resources must remain consistently utilized to maintain a smooth flow of work.

Work is continuously assigned to specific equipment (resources), and new work is released when capacity becomes available. As work is executed, feedback is generated when an oLPN (work) starts and completes processing. These feedback messages are

processed by Work Release to update work records and reflect newly available capacity. In e-commerce scenarios, the fulfillment process typically involves:

- A Unit Sorter and/or a Putwall to perform sorting operations.
- Creation of single oLPN totes that are packed at a pack station.

These components are considered critical resources and must be actively managed to ensure a continuous workflow. In some configurations, the Unit Sorter or Putwall may also perform the packing operation, eliminating the need for a separate packing step. As work progresses through these resources, feedback is sent to work release.

Unit Sorter

In a Unit Sorter flow, totes containing items for multiple partial orders are inducted into the sorter. The sorter directs inventory into chutes, where items are grouped by order until each order is fully sorted. An individual chute may hold multiple orders, typically limited to a configured maximum.

Once a chute is fully sorted, the inventory for those orders is either transferred to a new tote, or packed directly at the chute. If the inventory is transferred to a new tote (a multi-order tote), it is then routed to a Putwall for further processing. At the Putwall, one of two actions may occur:

- A second sort is performed to separate items into single-order oLPN totes, which are then sent to a pack station for final packing.
- Alternatively, the multi-order tote may be packed directly at the Putwall, depending on the configuration and flow design.

This flexible sorting and packing process ensures efficient handling of partial orders and supports high-throughput fulfillment operations.

Putwall

Multi-order totes can originate from a previous Unit Sorter operation, or direct picking of multiple orders brought to the Putwall. These totes are processed to create single-order oLPN totes, which are either packed directly at the putwall, or sent to a pack station for final packing.

In these fulfillment flows, the resource performing the initial sort or pack operation, such as a Chute or a Cubby, is typically designated as a critical resource. Maintaining continuous utilization of these resources is essential for achieving throughput goals. To manage the flow of work to these resources, feedback messages must be sent to Work Release when sorting or packing operations free up the resource. These messages indicate that capacity has been released and new work can be assigned. The work complete message will be sent either by the WCS or WM depending on the nature of operation.

The following matrix lists all the Work execution touch points, event details from these touch points, and corresponding updates on work release.

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Flow	Resource	Operation	Feedback Generation	Message Type & Content	Allocation Updates	Work Updates	Pack Wave Batch Updates
Unit Sorter	Unit Sorter	PACK or SORT	MHE Pack Cubed Directed MHE Pack from Tote (oLPN Packing from Tote) MHE Transit Tote Work Release gets the feedback message from the unit sorter to process and update the work status accordingly. The TIP message conveys the starting of the work and the CHUTE CLOSE message conveys the completion of message	TIP LPN,allocation keys,resource id, status, time stamp CHUTE CLOSE LPN,allocation keys,resource id, status, time stamp	On successful completion of WORK, update Allocation status to "WORK COMPLETED" for all the allocations associated with the WORK.	As part of the TIP message processing, update the WORK to "WORK STARTED" As part of the CHUTE CLOSE message processing, update the WORK to "WORK COMPLETED" Update the Resource on the WORK if they are different from what is already in the WORK as part of TIP or CHUTE CLOSE processing.	For first Work in the batch, as part of the TIP message processing, update the BATCH to "WORK STARTED" As part of the CHUTE CLOSE message processing for the last work in the batch, update the BATCH to "WORK COMPLETED"
Putwall	Putwall	PACK or SORT	MHE Pack from Tote MHE Transit Tote (Split Tote)	Line item status processing generates feedback from WM of Type " PACKING" Task ID, Allocation key, Time Stamp, Task Group	Allocation status updated to "WorkCompleted" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed		
				For oLPN Putwall, "PRINT" of oLPN label triggers the WORK COMPLETION by passing the list of Allocation Keys for the oLPN and RESOURCE		Update the WORK to "WORK COMPLETED" Update the Resource on the WORK if they are different from what is already in the WORK as sent in the message.	Update the BATCH to "WORK COMPLETED" for last work in the batch

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Flow	Resource	Operation	Feedback Generation	Message Type & Content	Allocation Updates	Work Updates	Pack Wave Batch Updates
			RF Pack from Tote (oLPN and Split Tote)	Feedback Type " PACKING" sent from WM Task ID, Allocation key, Time Stamp, Task Group	Allocation status updated to "WorkCompleted" if detail is completed "Cancelled" if the detail is cancelled Update the time stamps when the Packing is started and completed		
			RF Manage Putwall Or Pack from Tote with Putwall configured to Clear Cubby	CLEAR CUBBY triggering the message by passing the list of Allocation Keys for the oLPN and RESOURCE		Update the WORK to "WORK COMPLETED" Update the Resource on the WORK if they are different from what is already in the WORK as sent in the message.	Update the BATCH to "WORK COMPLETED" for last work in the batch
UI Pack Station	UI Pack Station	PACK		Once the oLPN goes to "Packed" status, work feedback is sent from WM by passing the list of Allocation Keys for the oLPN and RESOURCE	On successful completion of WORK, update Allocation status to "WORK COMPLETED" for all the allocations associated with the WORK.	Update the WORK to "WORK COMPLETED" Update the Resource on the WORK if they are different from what is already in the WORK as sent in the message.	
Replenishment	ASRS Decant	Decanting	Decant Message from WCS	Decant Message sent from WCS to WM by passing Container Number, Resource,Status,Time stamp	On successful completion of WORK, update Allocation status to "WORK COMPLETED" for all the allocations associated with the WORK.	Update the WORK to "WORK COMPLETED". Update the Resource on the WORK if they are different from what is already in the WORK as sent in the message.	

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Configuration

Scheduler for cleaning up HUNG work records is enabled and a Cleanup Age is defined in the scheduler workReleaseCleanupScheduler Job Schedule.

Business Updates

Work and corresponding allocations hanging are marked as Completed to free up available capacity.

Prioritization Manager

Prioritization in warehouse operations is a dynamic process. Priorities can change throughout the day due to various factors such as equipment availability, staffing levels, shipping plan changes, weather conditions, and carrier availability. To adapt to these changes, work release enables supervisors to re-prioritize tasks efficiently.

- **Real-Time Re-Prioritization:** Supervisors can adjust the priorities for pending work based on current conditions.
- **Rule-based Framework:** Priorities are determined using a configurable rule framework. Lower numerical values indicate higher priority.
- **HOT Allocations:** Allocations marked as HOT are treated with the highest urgency. The scoring algorithm accounts for the HOT flag when determining priority.

The process enables re-prioritization of outstanding allocations, which in turn updates the associated work. This helps balance workload across the warehouse and ensures optimal resource utilization.

Re-prioritization can be triggered in two ways:

- **Prioritization Scheduler**
  - Job Name: prioritizationSchedulerJobSchedule
  - Runs at a configured frequency to automatically update priorities
- **Work Visibility UI**
  - Allows manual intervention by supervisors to adjust priorities as needed.

Dynamic Prioritization in warehouse operations offers significant benefits by enabling real-time responsiveness to changing conditions such as staffing levels, equipment availability, and external disruptions like weather or carrier delays. By allowing supervisors to re-prioritize tasks based on a rule-driven framework, the system ensures that the most critical work is addressed first, improving order fulfillment accuracy and timeliness. The ability to mark allocations as HOT further enhances urgency handling, while automated and manual prioritization options provide flexibility. Overall, this approach helps balance workloads, optimize resource utilization, and maintain operational efficiency throughout the day.

Process Flow

This process outlines how the system evaluates and updates allocation priorities using configured rules and parameters.

1. Load Configuration Parameters
   - Retrieve the batch size value
   - Retrieve the prioritization time window
   - Load the prioritization strategy, including rule group and rule definitions
2. Fetch Eligible Allocations - Allocations are selected based on the following criteria:
   - The number of allocations matches the batch size retrieved in Step 1
   - The last prioritized timestamp of each allocation is older than the current prioritization time window
   - Allocation status is between 3000 [Template Assigned] and 7000 [Ready for Release]

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- Allocations must have a valid WORK_ID
- Allocations with work type oLPN or Order

3. Evaluate Prioritization Rules

   - Apply the rule framework to the filtered list of allocations.
   - Update the priority values based on rule evaluation outcomes.

Configuration

The various configuration that drives the re-prioritization process are:

1. Batch Size - This is a work release component property that decides how many allocations need to be processed for a scheduled run.
2. Prioritization Time Window - This is in work release parameters and used to determine the allocations that are past the time window interval
3. Define a Prioritization Strategy and associate the required rules to the strategy. On the rules define either a priority value or a hot flag.

Business Updates

The following updates are made as part of this process. For all matching evaluated allocations,

1. Update priority as below
   - If evaluated allocations returned as hot order
     - update priority = 0
   - else
     - update priority = configured priority value
   - endif
2. Update last prioritization to the current date time

Sample Examples

1. Re prioritize all pending Work shipping to Minnesota state as Priority 1
2. Re prioritize all pending Work shipping via FedEx as Priority 1
3. Re prioritize all pending Work past today's Cutoff as Priority 4
4. Re prioritize all pending VAS Work Shipping today as Priority 1

Replenishment Task Re-prioritization

Replenishment tasks that are released but not started yet can be dynamically re-prioritized. This ensures that high-priority pick/pack tasks are not delayed due to inventory shortages. By re-prioritizing replenishment tasks accordingly, the system helps maintain workflow continuity, supporting time order fulfillment and optimal resource usage. This is not applicable for Direct Task Creation mode and Replenishment Only waves. The Inventory constraint needs to be enabled.

To enable this, set the parameter “Do you want the system to reprioritize open replenishment tasks” in Work Release Parameters. When enabled, the system identifies open replenishment tasks that are associated with the pick/pack allocations selected by

the Work Release Scheduler. These allocations are chosen based on their Work Score, which reflects their priority. For each selected allocation, the system checks whether the to-be-picked quantity exceeds the on-hand quantity for the corresponding item-location combination. If replenishment is required, the system elevates the priority of the associated replenishment tasks to align with the score of the pick/pack allocation.

If the parameter is set to "0", the system determines the score and corresponding priority. If the parameter is set to a value that is not zero, you can manually assign a priority.

Re-optimization

Re-optimization is the process of re-evaluating allocations that were part of tasks not selected for release. It ensures that these allocations are either retained for future tasking or reassigned appropriately, based on system configuration and operational status. This is applicable only for non-bulk allocations.
When a task is not selected for release, the system applies a re-optimization strategy to determine how its associated allocations should be handled. Re-optimization is only for the allocations that are in "READY_FOR_TASKING" status. The behavior varies based on the configured Re-optimization level:

No Re-Optimization

- Allocations remain in the status "READY_FOR_TASKING".
- Decisions made by the Capacity Manager, such as inventory or resource assignments, are preserved.
- No further evaluation is performed.

Task Rule Manager

- If an allocation is not eligible for tasking, the existing task release template is nullified.
- Task criteria are re-evaluated to assign a new task release template and check if the allocations can be included in a future task.
- Capacity Manager decisions are retained.

Capacity Manager

- Allocations are moved to "`AWAITING_FOR_TEMPLATE_ASSIGNMENT`".
- Both task release template and work release template are re-evaluated.
- Previous decisions made by the Capacity Manager are discarded.

Pre-Capacity Manager

- Allocations are re-evaluated before capacity validation.
- Only work release template is re-evaluated.
- Caters to customers who prefer releasing tasks even if they are not optimized for efficiency or resource utilization.

For bulk allocations, the standard re-optimization levels do not apply. Instead, the system follows a simplified reset approach:

- All non-released bulk allocations, including both pick and pack, are reset to the status "`READY_FOR_CM`".
- Any pack work that is in a status less than "`RELEASED`" is also reset to "`READY_FOR_CM" status`.
- Similarly, non-selected pick work is moved back to "`READY_FOR_CM" status`.
- This ensures that bulk allocations are re-evaluated by the Capacity Manager and remain eligible for future task creation.

Re-Optimization is driven by the "Re-Optimization Level" configuration in the Work Release Parameters.

> Note: To use this feature, please contact your Manhattan representative.

Status Transition

Allocations, Work, and Tasks are the three transaction entities on which work release is based upon. As they move through planning and execution, the status of these entities changes to reflect their current stage. The status value plays an important role in determining the Resource WIP, Capacity, as well as Visibility. They control what actions can and cannot be carried out, serving as filters on the logic.

The status code and description represent the TO status, i.e., the status of the entity post successful execution of a step. It tracks the status transition of the entity as they move through work release. Once the work, allocation, and tasks have been released for execution, then the feedback moves the status further along. This movement is tracked in the feedback manager.

Allocation

Non Bulk

Step	Status Code	Description
Import Allocation	3000	READY_FOR_WORK_RELEASE
	3400	AWAITING_TEMPLATE_ASSIGNMENT
Determine Templates	3600 3800	TEMPLATE_ASSIGNMENT_FAILED OR TEMPLATE_ASSIGNED
Create Work	4000	READY_FOR_CM
Score Work		No Change
Capacity Manager Model		No Change
Post Processing	4400	READY_FOR_TASKING
Score Allocation		No Change
Create Tasks		No Change
Task Release Manager Model		No Change
Post Processing	4900	READY_FOR_RELEASE
	5000	RELEASED
Post Work Release - Status transition happens based on execution. See Feedback Manager for Details		
	5200	PICKING_STARTED
	5400	PICKING_COMPLETED
	5600	WORK_STARTED
	5800	WORK_COMPLETED
	9000	CANCELLED

Replenishment

	Status Code	Description
Import Allocation	3000	READY_FOR_WORK_RELEASE
	3400	AWAITING_TEMPLATE_ASSIGNMENT

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Determine Templates	3600 3800	TEMPLATE_ASSIGNMENT_FAILED OR TEMPLATE_ASSIGNED
Create Work	4000	READY_FOR_CM (If replenishment is not modeled for capacity, they go straight to ready for tasking)
Score Work		No Change
Capacity Manager Model		No Change
Post Processing	4400	READY_FOR_TASKING
Score Allocation		No Change
Create Tasks		No Change
Task Release Manager Model		No Change
Post Processing	4900	READY_FOR_RELEASE
	5000	RELEASED
Post Work Release - Status transition happens based on execution. See Feedback Manager for Details		
	5200	PICKING_STARTED
	5400	PICKING_COMPLETED
	5600	WORK_STARTED
	5800	WORK_COMPLETED
	9000	CANCELLED

Bulk

Bulk processing has two allocations - Pick and Pack.

PACK Allocations

Step	Status Code	Description
Import Allocation	3000	READY_FOR_WORK_RELEASE
	3400	AWAITING_TEMPLATE_ASSIGNMENT
Determine Templates	3600 3800	TEMPLATE_ASSIGNMENT_FAILED OR TEMPLATE_ASSIGNED
Create Work	4000	READY_FOR_CM
Score Work		No Change
Capacity Manager Model		No Change
Post Processing	4800	READY_FOR_DISTRIBUTION
Score Allocation		No Change
Create Tasks		No Change

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Task Release Manager Model		No Change
Post Processing	4900	READY_FOR_RELEASE
	5000	RELEASED
Post Work Release - Status transition happens based on execution. See Feedback Manager for Details		
	5200	PICKING_STARTED
	5400	PICKING_COMPLETED
	5600	WORK_STARTED
	5800	WORK_COMPLETED
	9000	CANCELLED

PICK Allocations

Step	Status Code	Description
Import Allocation	3000	READY_FOR_WORK_RELEASE
	3400	AWAITING_TEMPLATE_ASSIGNMENT
Determine Templates	3600 3800	TEMPLATE_ASSIGNMENT_FAILED OR TEMPLATE_ASSIGNED
Create Work	4000	READY_FOR_CM
Score Work		No Change
Capacity Manager Model		No Change
Post Processing	4800	READY_FOR_DISTRIBUTION
Score Allocation		No Change
Create Tasks		No Change
Task Release Manager Model		No Change
Post Processing	4900	READY_FOR_RELEASE
	5000	RELEASED
Post Work Release - Status transition happens based on execution. See Feedback Manager for Details		
	5200	PICKING_STARTED
	5400	PICKING_COMPLETED
	5600	WORK_STARTED
	5800	WORK_COMPLETED
	9000	CANCELLED

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Work

Work came into existence after "Create Work". From that point onwards, work mirrors the allocation status as described above. Do note that work is a group of allocations, and at any time, it may have allocations that are in different statuses. Work status reflects the maximum status of its allocation.

Work	Allocation	Status
W1	A1	READY_FOR_CM
	A2	RELEASED

Work W1 will be in a status of Released. As explained in the feedback manager, once the first allocation has been picked, work goes to work started, and once all the allocations have been picked, it goes to completed status. The UI status descriptions are customizable and can be adapted to implementation needs. The UI status descriptions are stored in the file "*label-miscellaneous.properties*".

Step	Status	Status Description	
Create Work	4000	READY_FOR_CM	Ready for Capacity Manager
Capacity Manager Model	4400	READY_FOR_TASKING	Ready for Tasking
	4800	READY_FOR_DISTRIBUTION	Ready for Distribution
Task Release for Execution	4900	READY_FOR_RELEASE	Ready for Release
	5000	RELEASED	Released
Post Work Release - Status transition happens based on execution. See Feedback Manager for Details	5200	PICKING_STARTED	Picking Started
	5400	PICKING_COMPLETED	Picking Completed
	5600	WORK_STARTED	Work Started
	5800	WORK_COMPLETED	Work Completed
	9000	CANCELED	Cancelled

Task

The tasks in work release are created in released status. Subsequent feedback moves the status further along.

Error Handling

Functional errors are logged with appropriate error codes. Below is a list of error codes and descriptions that are raised by work release.

Message Code	Component	Type	Message
WOR::000	WOR	ERROR	Mandatory Allocation Group(s) are missing
WOR::001	WOR	ERROR	Invalid Allocation ID list OR Query Criteria is missing
WOR::002	WOR	ERROR	Invalid Allocation ID list AND Query Criteria are not allowed together
WOR::003	WOR	ERROR	Resource Group(s) ID list AND Query Criteria are not allowed together
WOR::004	WOR	ERROR	Resource Group(s) AND Resource(s) are not allowed together
WOR::005	WOR	ERROR	PICK work record is not eligible for workTemplate assignment
WOR::006	WOR	ERROR	PACK work record is not eligible for taskTemplate assignment

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Message Code	Component	Type	Message
WOR::007	WOR	ERROR	Work record is not eligible for template assignment if work status is greater than READY_FOR_TASKING
WOR::008	WOR	ERROR	Work record is not found
WOR::009	WOR	ERROR	Both work and task templates are missing. Please provide one of them
WOR::010	WOR	ERROR	PACK Work record is not eligible for workTemplate assignment if work status is greater than READY_FOR_CM
WOR::011	WOR	ERROR	Work ID is missing
WOR::012	WOR	ERROR	Work {0} not eligible for priority change
WOR::013	WOR	ERROR	Allocation record is not found
WOR::101	WOR	ERROR	Import to Work Release Failed
WOR::118	WOR	ERROR	Task Release Template assignment failed for allocation {AllocationId}
WOR::138	WOR	ERROR	Number of allocations {Allocations} for slot template assignment completed
WOR::172	WOR	ERROR	Olpn dimension feasibility check failed for Allocation
WOR::175	WOR	ERROR	Tote dimension feasibility check failed for Allocation
WOR::177	WOR	ERROR	Template Assignment Failed for one of this Order Allocation
WOR::183	WOR	ERROR	Eligible task group not available
WOR::184	WOR	ERROR	Task capacity not available
WOR::187	WOR	ERROR	Work release allocations not submitted for task creation due to missing template count [{TemplateNotAssignedCount}]
WOR::226	WOR	ERROR	Template Assignment Failed for one of the allocations in this oLPN

Exception Handling

If the work release allocations have incorrect data, the allocations are validated against certain data conditions and moved to a new "Processing Error" status to avoid halting the work release scheduler run. The work release allocations that are moved to "Processing Error" status are not evaluated for further processing by the scheduler. Additionally, these allocations can be viewed in the Pending Work Release Allocations Inquiry screen. The error description is updated on the allocation with the reason for failure.

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